

FINITE FAILURE SCHEMES AND THE RECONSTRUCTION OF QUADRATIC PENCILS

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ABSTRACT. Every quadratic pencil in dimension $n \geq 3$ is determined up to projective equivalence by the abstract $(n^2 + 2n - 4)$ -th neighbourhood of the socle Grassmannian in its multiplication-failure scheme. Every lower-order neighbourhood is independent of the pencil. We prove this by recovering the oriented tensor cone and its full Pluecker coefficient line intrinsically. The first relation spaces embed the entire pencil parameter scheme and give universal finite neighbourhoods compatible with arbitrary complex base change. For regular pencils they recover spectral sheaves, their Fitting incidence schemes, and the reciprocal critical correspondence. At fixed discriminant and reduced rank data, the finite invariant realizes and distinguishes every spectral dominance transition. For definite real pencils, spectral projectors give a congruence-covariant open family of data on which the critical correspondence is finite etale and all its points are real, simple, and globally labelled. For split semisimple data, the projective critical scheme is finite flat through collisions, with Hessian ramification and a trace discriminant. The full web, contraction, polarized, and boundary theory is retained in the accompanying supplement.

1. INTRODUCTION

A degeneracy locus usually forgets the map that defines it. We ask whether its nonreduced scheme structure can recover that map, without an ambient embedding or a marking of the normal directions. For the multiplication-failure schemes of quadratic pencils the answer is exact: there is one uniform infinitesimal order at which the entire pencil first becomes visible. The inverse theorem applies to every pencil, including singular pencils. Its regular locus then carries the spectral and critical correspondences studied below.

1.1. The principal theorem. Let V be a complex vector space of dimension $n \geq 3$, and set

$$W = \mathrm{Sym}^2 V, \quad N = \binom{n+1}{2}, \quad R \in \mathrm{Gr}(2, W), \quad S_R = W/R, \quad p = N - 2.$$

The algebra $B_R = \mathbb{C} \oplus V \oplus S_R$ has multiplication $\mathrm{Sym}^2 V \twoheadrightarrow S_R$, with $(V \oplus S_R)S_R = 0$. On $X_R = \mathrm{Gr}(n, V \oplus S_R)$ let $\mathcal{K}$ denote the tautological bundle. For any $m \geq 2$, define

$$\widehat{D}_R = V(\mathrm{Fitt}_0 \mathrm{coker}[\mathrm{Sym}^m(\mathcal{O} \oplus \mathcal{K}) \longrightarrow B_R \otimes \mathcal{O}]). \quad (1.1)$$

This is the literal maximal-minor scheme of the multiplication map; it is independent of $m \geq 2$. Its reduction is the fixed Schubert divisor where $\mathcal{K} \rightarrow V$ drops rank. Let $B = \mathrm{Gr}(n, S_R)$, let $\mathfrak{b}_R$ be its ideal in $\widehat{D}_R$, and put

$$Z_R^{[k]} = V(\mathfrak{b}_R^{k+1}) \subset \widehat{D}_R, \quad d = n + 2p = n^2 + 2n - 4.$$

The neighbourhood is taken along the whole projective socle Grassmannian, not at one framed affine point.

Date: September 24, 2026.

2020 Mathematics Subject Classification. 14M12, 14B05, 14D20, 13C40, 62R01.

Key words and phrases. Intrinsic reconstruction, quadratic pencils, Fitting schemes, finite neighbourhoods, spectral sheaves, likelihood correspondence.

Theorem 1.1 (Sharp intrinsic finite reconstruction). *For fixed $n \geq 3$, the following assertions hold on all of $\mathrm{Gr}(2, \mathrm{Sym}^2 V)$.*

(i) *For every pair of pencils R, R' ,*

$$Z_R^{[d]} \simeq Z_{R'}^{[d]} \iff R' = gR \text{ for some } g \in \mathrm{PGL}(V).$$

The isomorphism of neighbourhoods is abstract and unmarked. For each $0 \leq k < d$, the isomorphism class of $Z_R^{[k]}$ is independent of R . Thus d is the smallest uniform reconstruction order.

(ii) *In fixed tensor coordinates, the first relation spaces define a closed immersion*

$$\mathrm{Gr}(2, W) \longrightarrow \mathrm{Gr}\left(\binom{N}{2}, \mathrm{Sym}^d \mathrm{Hom}(U, V)^*\right), \quad \dim U = n.$$

This assertion includes infinitesimal parameter schemes and singular pencils.

(iii) *These relations glue to the actual universal order- d neighbourhood over the relative socle Grassmannian. It is finite flat there of rank*

$$\binom{n^2 + d}{d} - \binom{N}{2},$$

projective and flat over the pencil parameter scheme, and compatible with arbitrary complex base change with the relative socle retained.

Theorem 5.1 proves (i); Theorems 5.3 and 7.1 prove (ii)–(iii). The distinction between their inputs is essential: (i) uses an abstract global scheme, whereas (ii)–(iii) retain the relative socle and a tensor presentation. No equivalence of unmarked moduli stacks is asserted.

Here is the mechanism behind the theorem. In graph coordinates $T : \mathcal{U} \rightarrow V$, where $\mathcal{U}$ is tautological on B , the homogeneous ideal is

$$(\det T)I_p(\gamma_R \mathrm{Sym}^2 T), \quad \gamma_R : W \twoheadrightarrow S_R.$$

It is generated in degree d . The abstract finite scheme recovers B , its degree-one conormal, and the first relation module. The reduced determinant cone has two rank-one rulings. Only its $\mathbb{P}(V)$ -ruling is projectively trivial over B ; a Schubert line distinguishes the other one. Regular descent therefore recovers a single constant projective coordinate transformation on V , rather than unrelated changes for different coefficients. Cancelling the determinant line then recovers the residual coefficient module. For pencils its exterior representation is irreducible; exterior duality identifies its coefficient line with the full Pluecker line of R . The first visible degree alone is formal; these intrinsic identifications are the substantive inverse argument.

1.2. Spectral and critical correspondences. On the regular locus, the reconstructed pencil determines its torsion spectral sheaf, not merely its determinant. Theorems 6.3 and 7.4 recover both the elementary divisors and their scheme-theoretic Fitting incidence loci. Theorem 8.1 applies this to every fixed discriminant and reduced-rank stratum: the classical dominance specializations are realized by flat families of the finite neighbourhoods, and every strict change first becomes distinguishable at order d . The qualifier *reduced* matters; the full spectral Fitting schemes need not remain fixed.

There is a further functorial output. The reciprocal critical correspondence is the zero scheme of the two score equations on the invertible members of a pencil, with the data matrix allowed to vary. Theorem 10.1 recovers its isomorphism class from the finite failure invariant and constructs it in families. Thus the passage from finite failure to likelihood has a specified algebraic intermediate object, rather than a parallel numerical analogy. For a supplied definite real realization, Theorem 10.2 gives a congruence-covariant open family of data, a finite etale

critical algebra of rank $2r - 3$, and globally labelled real analytic critical sections. Its residue and separated-root argument builds on the complete proof of Theorem 9.1. That latter theorem proves the assertion formulated in [40, Conjecture 4.5], for unrestricted real data and arbitrary eigenvalue multiplicities. Neither a real form nor a definite member is inferred from an unmarked complex scheme, and positivity of the data or of all critical points is not asserted.

The affine etale description is not the end of the correspondence. Theorems 11.1 and 11.2 retain the full projective critical divisor on the split semisimple spectral-data open. It is finite locally free of degree $2r - 3$, including critical collisions and directions at infinity. Its Hessian-degeneracy ideal is its relative ramification ideal, and its trace discriminant records collision multiplicities. Proposition 11.3 exhibits a length-three nonreduced critical fibre in one fixed pencil. This continues the same reconstruction chain without changing the hypotheses of the all-pencil inverse theorem.

1.3. Proof dependencies and further results. The principal argument is independent of exterior recombination. The following table specifies the logical inputs; the implication from reconstructed matrices to the critical correspondence is formal, whereas its totally real open family requires an additional proof.

Step	Input	Output
Intrinsic coefficients	Literal Fitting ideal; oriented rank-one rulings	One projective frame and the full Pluecker line
Finite reconstruction	Coefficient inverse; single relation degree	Sharp uniform order; parameter immersion
Relative spectral read-out	Glued finite neighbourhood; regular pencil	Spectral sheaf and Fitting incidence schemes
Critical correspondence	Reconstructed pencil; universal score equations	Congruence-covariant algebraic correspondence
Projective critical divisor	Split semisimple data; disjoint simple poles	Finite flat scheme; ramification and trace discriminant
Totally real cover	Supplied real definite realization; residues and signs	Finite etale algebra and labelled real critical sections

The supplement retains every web, contraction, polarized, polar, and boundary theorem with its full proof. In particular, Theorem S1.1 recovers all basepoint-free four-dimensional webs with smooth Jacobian quartic; the all-ranks contraction theorem and exact Jacobian–Casimir factorization supply its recombination mechanism. They are substantial further results, not hidden hypotheses of the pencil theorem. The first-layer K3 construction, hyperelliptic interpretation, component-fibre classification, and primary boundary calculations remain available at their original level of strength.

1.4. Relation to classical results. The Weierstrass–Segre classification, its orbit closures, the harmonic decomposition, and exterior-support bounds are classical inputs, identified at the statements where they are used. In particular, [40, Theorem 1.1 and Corollary 2.1] classify regular pencils by their labelled elementary divisors; [44, equation (2)] gives the selfadjoint orthogonal orbit calculation underlying the fixed-spectrum discussion. The inverse problem here starts instead with an unmarked failure scheme. The exact differential dictionary for the auxiliary contraction is given in Section S12; it separates classical harmonic tools from the normalization of the specified map.

Ballico’s later paper [3, Theorem 0.2; §1, Remark 1.2 and equation (6)] detects failure of quadratic normality on finite linear sections of a surface using a relative multiplication map. Our parameter is instead the quotient algebra B_R , and our output is reconstruction from an

unmarked infinitesimal failure scheme. The inverse argument must be compared with that relative construction, not merely with its terminology.

Ballico's paper [2] is a directly relevant predecessor. Its complete text has not been obtained for the present comparison. Consequently we do not infer nonanticipation from its title, first page, or later citations. The six theorem-level comparisons requiring that text are recorded explicitly in the accompanying referee response. This documentary limitation is separate from the hypotheses and proofs of the theorems stated here.

2. INTRINSIC TENSOR RULINGS

The first step of the inverse problem is to recognize the two tensor factors of a determinant cone without an ambient marking. The following scheme-theoretic ruling calculation and descent lemma supply precisely this step. They apply to the pencil argument independently of the web and exterior-contraction constructions in the supplement.

Lemma 2.1 (The relative scheme of maximal rank-one spaces). *Let B be any complex scheme and E, F vector bundles of rank $n \geq 2$. In $\mathrm{Gr}_B(n, E \otimes F)$, let $Z(E, F)$ be the zero scheme of the restriction of the quadratic rank-one equations to the universal subbundle $\mathcal{T}$. Equivalently it is the zero scheme of*

$$\mathrm{Sym}^2 \mathcal{T} \longrightarrow \bigwedge^2 E \otimes \bigwedge^2 F.$$

The natural morphism

$$\mathbb{P}_B(E) \sqcup \mathbb{P}_B(F) \longrightarrow Z(E, F), \quad \ell \mapsto \ell \otimes F, \quad m \mapsto E \otimes m, \quad (2.1)$$

is an isomorphism. It commutes with arbitrary change of complex base, including change to a nonreduced base.

Proof. We first work over $\mathbb{C}$. Each displayed map to the Grassmannian is a closed immersion. Its composition with the Pluecker embedding is the degree- n Veronese embedding on $\ell^{\otimes n} \otimes \det F$, respectively $\det E \otimes m^{\otimes n}$, followed by its natural linear inclusion into $\bigwedge^n(E \otimes F)$. This linear inclusion is injective: polarization of the nonzero pure-power map gives the corresponding irreducible symmetric-power summand. The rank-one equations vanish identically on both universal families, so the closed immersions factor through $Z(E, F)$.

Every geometric point is in one of the two images. Indeed the sum of two rank-one tensors is rank at most one only if their first factors or their second factors are proportional. Starting with two independent tensors, this observation forces a linear space consisting of rank-one tensors to fix the same factor throughout. A space of dimension n is then exactly one of the displayed spaces. The two images are disjoint: a space fixing both factors has dimension one, not n .

We verify that these pointwise statements omit no infinitesimal structure. At $\ell \otimes F$, choose a generator of ℓ and a complement in E . A Grassmannian tangent vector is a map $\phi : F \rightarrow (E/\ell) \otimes F$. For each coordinate of E/ℓ , write its component as $A : F \rightarrow F$. Linearizing the restricted minors gives

$$u \wedge A(u) = 0 \quad \text{for every } u \in F.$$

Polarizing this diagonal identity gives

$$u \wedge A(v) + v \wedge A(u) = 0 \quad (u, v \in F).$$

In characteristic zero the diagonal and polarized identities are equivalent. These are precisely all coefficients of the bilinear linearization of the restricted quadratic map on $\mathrm{Sym}^2 F$, so no infinitesimal equation is omitted by testing the diagonal. Taking the basis vectors shows that A is diagonal; taking their pairwise sums makes all its diagonal entries equal. Conversely a scalar

A satisfies all the linearized equations. Thus the Zariski tangent space to $Z(E, F)$ is precisely $\text{Hom}(\ell, E/\ell)$, of dimension $n - 1$. The other image has the symmetric calculation.

At every closed point the closed immersed projective space gives local dimension at least $n - 1$, whereas the tangent calculation gives embedding dimension exactly $n - 1$. The local ring is therefore regular. A finite-type complex scheme whose closed-point local rings are regular is reduced: a nonempty support of its nilradical would contain a closed point. In particular there is no nilpotent thickening or embedded associated structure at either image. The disjoint union of the two closed immersions is itself a closed immersion, surjective on geometric points. Its ideal in the now reduced target vanishes at every closed point, so it is zero. This proves the scheme isomorphism over $\mathbb{C}$.

For general B , trivialize E, F on a common open cover. The Grassmannian and the zero scheme of the displayed universal quadratic map are, on such an open, the base change of the constant construction just proved. Thus (2.1) is an isomorphism there even if the base has nilpotents. The maps are natural under changes of frame and hence glue. This argument is an identification by equations and base change, not an inference of relative flatness from a fibrewise tangent calculation. $\square$

Lemma 2.2 (Oriented relative Segre descent). *Let B be a connected projective complex variety with $H^0(B, \mathcal{O}_B) = \mathbb{C}$, and let V, V' have the same dimension $n \geq 2$. Let $\mathcal{U}, \mathcal{U}'$ be vector bundles of rank n on B . Suppose a linear bundle isomorphism*

$$\alpha : V \otimes \mathcal{U}^* \xrightarrow{\sim} V' \otimes \mathcal{U}'^*$$

preserves the rank-one cones and the ruling whose parameter bundle is $\mathbb{P}(V) \times B$. The induced isomorphism $\mathbb{P}(V) \times B \simeq \mathbb{P}(V') \times B$ is regular and is induced by one constant projective isomorphism $[g] : \mathbb{P}(V) \rightarrow \mathbb{P}(V')$. Locally $\alpha(T) = gTh^{-1}$, with the same $[g]$ on every chart. The possible line-bundle twists of vector-bundle lifts do not change the projective left coefficient lines of any homogeneous functorial coefficient construction.

Proof. Lemma 2.1, applied to $E = V \otimes \mathcal{O}_B$ and $F = \mathcal{U}^*$, identifies the closed relative Fano scheme with $(\mathbb{P}(V) \times B) \sqcup \mathbb{P}_B(\mathcal{U}^*)$. In particular its two components carry their displayed scheme structures, with no additional nilpotent or embedded components. The isomorphism α induces a regular map of these relative parameter schemes, and the orientation hypothesis preserves the first component.

For completeness, an isomorphism of two trivial $\mathbb{P}^{n-1}$ -bundles over any base gives a regular projective-matrix-valued map, not merely a fibrewise collection. Its pullback of the relative $\mathcal{O}(1)$ has the form $\mathcal{O}(1) \otimes p^*L$ for a line bundle L on the base. To see this, tensor by $\mathcal{O}(-1)$; the resulting line bundle is trivial on every projective-space fibre, and its pushforward and evaluation identify it with a pullback from the base. Pushing forward $\mathcal{O}(1)$ now gives an isomorphism of the corresponding trivial rank- n bundles, with that line-bundle twist. On charts trivializing L , it is an invertible matrix g_i of regular functions. On overlaps, $g_j = u_{ij}g_i$ for an invertible regular function u_{ij} . The classes $[g_i]$ thus glue to a regular morphism to the projective-isomorphism scheme, identified after choosing fixed bases with PGL_n . This accounts explicitly for the possible absence of a global linear lift.

The group PGL_n is affine: it is the determinant-nonzero principal open in the projective space of matrices. A morphism from B to an affine scheme factors through $\text{Spec } H^0(B, \mathcal{O}_B) = \text{Spec } \mathbb{C}$, so the projective map is constant. Choose a constant linear lift g . Each local g_i then differs from g by a scalar unit. After removing its two projective-factor actions, a linear map preserving every decomposable line is scalar: apply it to sums of two tensors with one common factor. Consequently the remaining action is a right bundle map, and the scalar units may be absorbed in its local lifts h . This proves the local tensor formula.

Finally a scalar multiplying a lift acts on a homogeneous functor of degree a by its scalar to the power a (or $-a$ on its dual). It cannot alter a projective coefficient line. For the degree-twelve construction below, both left Cauchy factors have the same degree twelve; the exterior-duality determinant multiplier is common as well. Thus neither line-bundle transition functions nor right changes of frame can introduce two different projective transformations on the left. $\square$

3. AN INTRINSIC COEFFICIENT PRINCIPLE

The web problem combines two operations: extracting coefficient data from an abstract scheme and reconstructing a relation space from those data. The first operation admits a formulation independent of webs or quadratic multiplication. We state the criterion and retain its polar-restriction application for linear systems in every dimension. Theorem 4.1 subsequently derives all of its geometric hypotheses directly for a full Grassmannian family of the original quadratic multiplication schemes.

3.1. The structural criterion. Let X be a finite-type complex scheme with a closed subvariety B specified by a construction preserved by every abstract scheme isomorphism. Write $\mathfrak{b} \subset \mathcal{O}_X$ for its ideal and

$$\mathcal{G} = \bigoplus_{r \geq 0} \mathfrak{b}^r / \mathfrak{b}^{r+1}, \quad \mathcal{A} = \text{Sym}_{\mathcal{O}_B}(\mathcal{G}_1), \quad \mathcal{I} = \ker(\mathcal{A} \twoheadrightarrow \mathcal{G}).$$

Assume the following conditions, for fixed integers $n \geq 2, m \geq 1$.

- (i) B is connected and projective, $H^0(B, \mathcal{O}_B) = \mathbb{C}$, and $\mathcal{G}_1$ is locally free. Its dual admits a presentation $V \otimes \mathcal{U}^*$, where $\dim V = \text{rank } \mathcal{U} = n$.
- (ii) The reduction of $\text{Spec}_B \mathcal{G}$ in its degree-one ambient bundle is the generic determinant cone. The bundle $\mathbb{P}_B(\mathcal{U}^*)$ is not projectively trivial. Thus precisely one of the two ruling parameter bundles of its rank-one locus is projectively trivial.
- (iii) If $\mathcal{D} \subset \mathcal{A}$ is the ideal of that reduced cone and $\mathcal{L} = \mathcal{D}_n$, then $\mathcal{I} = \mathcal{D}\mathcal{J}$, where $\mathcal{J}$ is generated in degree m . Under the Cauchy decomposition in any local frame of $\mathcal{U}$,

$$\mathcal{J}_m = \bigoplus_{\lambda \in \Lambda} L_\lambda \otimes \mathbb{S}_\lambda \mathcal{U} \subset \text{Sym}^m(V^* \otimes \mathcal{U}), \quad L_\lambda \subset \mathbb{S}_\lambda V^*, \quad (3.1)$$

where Λ is a specified collection of partitions of m of length at most n , and the left subspaces are constant over B . Zero subspaces are permitted. The Cauchy inclusions, not just their representation labels, are the coefficient realizations $\beta \otimes \xi \mapsto [T \mapsto \beta(\mathbb{S}_\lambda(T)\xi)]$.

These conditions are hypotheses on the normal cone and its residual coefficient module. They do not assume that an abstract isomorphism extends to an ambient space, preserves a chosen frame, or already acts on all the left subspaces through a common projective transformation.

Theorem 3.1 (Intrinsic coefficient reconstruction). *Suppose X, B satisfy (i)–(iii). The abstract scheme X determines the collection $(L_\lambda)_{\lambda \in \Lambda}$ up to one common projective coordinate transformation on V . More precisely, an isomorphism between two such schemes induces a constant projective isomorphism $[g] : \mathbb{P}(V) \rightarrow \mathbb{P}(V')$ such that*

$$L_\lambda = \mathbb{S}_\lambda(g)^* L'_\lambda \quad (\lambda \in \Lambda).$$

The equality means equality of subspaces; scalar changes of a linear lift of $[g]$ do not affect it.

Proof. An isomorphism transports B , every power of $\mathfrak{b}$, $\mathcal{G}_1$, and the surjection $\mathcal{A} \twoheadrightarrow \mathcal{G}$. In particular it transports $\mathcal{I}$ in the intrinsic degree-one ambient bundle. Iterated reduced singular loci of the determinant cone give its rank-one locus. Lemma 2.1 reconstructs its two ruling parameter schemes, including their scheme structure. Condition (ii) distinguishes the trivial ruling intrinsically. Lemma 2.2 then gives one constant projective map $[g]$. After choosing

a constant linear lift and local frames on both bases, the ambient bundle map has the form $T \mapsto gTh^{-1}$. These local choices are made only after constructing the graded algebra, its degree-one part, and the ruling.

Locally $\mathcal{D} = (d)$ with $d = \det T$. Its leading coefficient is a unit, so it is a nonzerodivisor over every coefficient ring. Hence $(\mathcal{I} : \mathcal{D}) = \mathcal{J}$, and

$$\mathcal{J}_m \simeq \mathcal{L}^* \otimes \mathcal{I}_{n+m}.$$

This recovers the residual module without choosing a scalar determinant equation. Under $T \mapsto gTh^{-1}$, the Cauchy coefficient map transports a tensor by $\mathbb{S}_\lambda(g)^* \otimes \mathbb{S}_\lambda(h^{-1})$. Projection to the specified Cauchy summand, followed by contraction against its whole right dual, therefore transports exactly L'_λ to L_λ . The right map is invertible and has no effect on that left image. Both determinant-line transition functions and changes of a lift of $[g]$ act by scalar units. Since every partition has size m , this ambiguity is common in degree and disappears on subspaces. This proves the assertion. $\square$

4. QUADRATIC RELATION SPACES AND INTRINSIC RECONSTRUCTION OF PENCILS

The multiplication construction itself supplies a wider natural class for the coefficient criterion. In this section the number of quadratic relations is allowed to vary independently of the embedding dimension. For two relations the resulting theorem recovers every pencil of quadrics, including singular pencils. No extra block is added to the multiplication map.

4.1. Automatic coefficient extraction for quadratic multiplication. Let $\dim V = n \geq 3$, $W = \text{Sym}^2 V$, $N = n(n+1)/2$, and $R \in \text{Gr}(r, W)$, where $1 \leq r < N - n$. Put $p = N - r > n$, $S = W/R$, and

$$A_R = \mathbb{C} \oplus V \oplus S, \quad X_R = \text{Gr}(n, V \oplus S).$$

The only nonzero product in the maximal ideal is the quotient map $\gamma : \text{Sym}^2 V \rightarrow S$. Write $\mathcal{K}$ for the tautological bundle on X_R , and use the same definition (1.1) of $\widehat{D}_R$, with A_R in place of B_R . Thus $\widehat{D}_R$ is the failure scheme of generating A_R by a varying n -plane and the unit. The notation in this section agrees with the web construction when $(n, r) = (4, 4)$.

Let $\Lambda_{n,p}$ denote the partitions occurring in the classical multiplicity-free decomposition

$$F(V) := \bigwedge^p \text{Sym}^2 V = \bigoplus_{\lambda \in \Lambda_{n,p}} \mathbb{S}_\lambda V. \quad (4.1)$$

Every partition has size $2p$; terms of length greater than n are omitted. Choose a nonzero volume form on S and let $\beta_R \in F(V)^*$ be its pullback by $\bigwedge^p \gamma$. Its component in $\mathbb{S}_\lambda V^*$ is denoted by $\beta_{R,\lambda}$. Only the lines of the nonzero components will be used, so the volume choice is immaterial.

Theorem 4.1 (Automatic intrinsic coefficient extraction). *For every $R \in \text{Gr}(r, \text{Sym}^2 V)$ in the stated range, the abstract scheme $\widehat{D}_R$ determines*

$$(\mathbb{C}\beta_{R,\lambda})_{\lambda \in \Lambda_{n,p}}$$

up to one common projective transformation of V , with zero components recorded as zero subspaces. More precisely, the intrinsic construction recovers the base $B = \text{Gr}(n, S)$, its degree-one normal-cone bundle, and the residual ideal generated in degree $2p$ with coefficient module

$$\mathcal{J}_{2p} = \bigoplus_{\lambda \in \Lambda_{n,p}} \mathbb{C}\beta_{R,\lambda} \otimes \mathbb{S}_\lambda \mathcal{U} \subset \text{Sym}^{2p}(V^* \otimes \mathcal{U}), \quad (4.2)$$

where $\mathcal{U}$ is the tautological rank- n bundle on B . Consequently all geometric hypotheses of Theorem 3.1 hold for this entire Grassmannian family, not just for smooth webs.

Proof. On a frame chart of $\mathcal{K}$, write its inclusion as $\binom{M}{A_0}$, with M an $n \times n$ matrix. Since A_R is cube-zero, columns of degree greater than two vanish. After retaining the scalar, linear, and quadratic columns, multiplication has the matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & M & 0 \\ 0 & A_0 & \gamma \operatorname{Sym}^2 M \end{pmatrix}. \quad (4.3)$$

A nonzero maximal minor must use the scalar column and all n linear columns: these are the only columns meeting the first $1+n$ rows. Expanding along those rows gives the identity of ideals

$$I_{\widehat{D}_R} = (\det M) I_p(\gamma \operatorname{Sym}^2 M). \quad (4.4)$$

The same expansion works for every $m \geq 2$. Where M is invertible, the quadratic map is surjective. Hence the reduction of $\widehat{D}_R$ is exactly the Schubert divisor where M is singular, independent of R after identifying the vector spaces S .

Successive reduced singular loci of that divisor are the projection rank loci. Their last nonempty member is $B = \operatorname{Gr}(n, S)$. For clarity, the usual determinantal assertion here is local: at a projection-rank- k point, an invertible k -minor splits away, leaving a generic square $(n-k)$ -matrix and smooth coordinates. Every remaining entry can vary independently by moving the kernel of the projection in V modulo its image. Thus its reduced singular locus is the next rank locus. Since $p > n$, the final base is nonempty, connected and projective, with $H^0(B, \mathcal{O}_B) = \mathbb{C}$.

Near B , a Grassmannian chart consists of a varying plane in S and its graph $T : \mathcal{U} \rightarrow V$. Formula (4.4) is independent of the other graph coordinates and homogeneous in T . It therefore gives its actual normal-cone ideal, not merely an initial reduced support:

$$C_B \widehat{D}_R \subset \operatorname{Tot} \operatorname{Hom}(\mathcal{U}, V), \quad \mathcal{I} = (\det T) \mathcal{J}, \quad \mathcal{J} = I_p(\gamma \operatorname{Sym}^2 T). \quad (4.5)$$

There are no degree-one equations. The ambient bundle in (4.5) is consequently the dual of the intrinsically recovered degree-one conormal. Its reduced ideal is the determinant ideal $\mathcal{D}$. The determinant polynomial is monic for a suitable monomial order, so cancellation gives $(\mathcal{I} : \mathcal{D}) = \mathcal{J}$. This is an intrinsic ideal quotient; choosing a generator of the determinant line is not part of the recovered datum.

The rank-one locus of the reduced cone determines the two ruling parameter schemes by Lemma 2.1. They are $\mathbb{P}(V) \times B$ and $\mathbb{P}_B(\mathcal{U}^*)$. Exactly the first is projectively trivial. Indeed a Schubert line in $\operatorname{Gr}(n, S)$ restricts $\mathcal{U}^*$ to $\mathcal{O}(1) \oplus \mathcal{O}^{n-1}$. A trivial projective bundle on that line would make this bundle a line-bundle twist of $\mathcal{O}^n$; its determinant would then have degree divisible by n , rather than degree one. Thus the trivial ruling is intrinsically oriented. Lemma 2.2 applies: an abstract isomorphism gives a morphism $B \rightarrow \operatorname{PGL}(V)$, constant because B is connected projective and $\operatorname{PGL}(V)$ is affine. Local linear lifts and right changes of frame give $T \mapsto gTh^{-1}$ for this one fixed $[g]$.

It remains to prove (4.2), rather than assume it as a condition on the example. The minors are the image, for fixed β_R , of the universal coefficient map

$$F(V)^* \otimes F(U) \longrightarrow \operatorname{Sym}^{2p}(V^* \otimes U), \quad \beta \otimes \xi \longmapsto [T \mapsto \beta(F(T)\xi)]. \quad (4.6)$$

In characteristic zero, the polynomial functor F has the multiplicity-free decomposition (4.1); see [37, Theorem 3.5]. The Cauchy decomposition of the target contains each diagonal $\mathbb{S}_\lambda V^* \otimes \mathbb{S}_\lambda U$ once, and no off-diagonal pair. Equivariance kills every off-diagonal pair. On each diagonal pair the coefficient map is nonzero: identify U with V , evaluate at $T = 1$, and choose β, ξ in that summand with $\beta(\xi) \neq 0$. Schur's lemma then identifies its image with the specified Cauchy summand. This is the actual matrix-coefficient inclusion, not merely a dimension count. Keeping β_R fixed proves (4.2). All generators have degree $2p$, and at least one minor is nonzero since γ is surjective.

Under $T \mapsto gTh^{-1}$, the right map acts within the full factor $\mathbb{S}_\lambda U$. Contracting against its dual therefore leaves exactly the left coefficient line transformed by $\mathbb{S}_\lambda(g)^*$. Determinant-line scalars and local lift ambiguities do not change these lines. This proves the theorem. $\square$

The theorem does not assert that separately projectivized Schur components always determine their relative scalars. For webs this is the recombination problem solved in the supplement. For pencils that issue disappears altogether.

4.2. Every pencil, without a regularity hypothesis.

Lemma 4.2 (The exterior square of the quadratic representation). *For $n \geq 2$, $\bigwedge^2 \text{Sym}^2 V = \mathbb{S}_{(3,1)} V$ is irreducible. Consequently $\bigwedge^{N-2} \text{Sym}^2 V$ is irreducible as well.*

Proof. The vector $x_1^2 \wedge x_1 x_2$ is a highest-weight vector of weight $(3, 1)$. By complete reducibility it generates a copy of $\mathbb{S}_{(3,1)} V$. The hook formula gives

$$\dim \mathbb{S}_{(3,1)} V = \frac{n(n-1)(n+1)(n+2)}{8} = \binom{N}{2}.$$

This exhausts the exterior square. The last assertion follows by exterior duality. These are classical representation identities. $\square$

Theorem 4.3 (Intrinsic reconstruction of all quadratic pencils). *Let $n \geq 3$ and let R, R' be two-dimensional subspaces of $\text{Sym}^2 V$. For the natural multiplication-failure schemes of their cube-zero algebras of Hilbert function $(1, n, N-2)$,*

$$\widehat{D}_R \simeq \widehat{D}_{R'} \iff R' = gR \text{ for some } g \in \text{PGL}(V). \quad (4.7)$$

No pencil, base locus, or discriminant is required to be smooth or regular. The isomorphism on the left is abstract and carries no supplied ambient embedding, base, or tensor-factor marking. All the reduced schemes are isomorphic to the same Schubert divisor.

Proof. Here $p = N - 2 > n$. By Theorem 4.1 and Lemma 4.2, the normal cone has only one left coefficient line to recover: the entire line $\mathbb{C}\beta_R$. The canonical duality

$$\bigwedge^{N-2} W^* \simeq \bigwedge^2 W \otimes (\det W)^* = \bigwedge^2 W \otimes (\det V)^{-(n+1)} \quad (4.8)$$

sends that line to the Pluecker line of R , with the same projective transformation g . Indeed the pullback of a quotient volume is the annihilator-volume dual of the kernel R . The common determinant character is a scalar on projective space. A nonzero decomposable two-vector recovers its plane as $\{w \in W : w \wedge w_R = 0\}$. Thus an abstract scheme isomorphism implies $R' = gR$.

Conversely such a linear coordinate change induces the quotient isomorphism $S_R \rightarrow S_{R'}$, the algebra isomorphism $A_R \rightarrow A_{R'}$, and the corresponding Grassmannian isomorphism. It intertwines multiplication and hence preserves its zeroth Fitting ideal. This proves the reverse implication. The assertion about reductions was proved in (4.4). $\square$

4.3. Discriminants, elementary divisors, and hyperelliptic moduli. A regular pencil means one containing an invertible symmetric matrix; it need not have a reduced discriminant. The classical invariant of such a pencil consists of its discriminant points together with their Jordan-block partitions, usually called its Weierstrass–Segre data. We use the root-labelled partitions, not a pairing of an unlabelled list with an independently unlabelled divisor. Their classification is classical; compare [40, Theorem 1.1, Corollary 2.1 and §2]. The following consequence identifies precisely what the intrinsic failure scheme remembers beyond the discriminant.

Corollary 4.4 (Intrinsic recovery of elementary divisors). *For a regular pencil in any dimension $n \geq 3$, the abstract scheme $\widehat{D}_R$ determines all root-labelled elementary divisors of a matrix pencil, up to projective reparametrization of its parameter line. Conversely these data determine its abstract failure scheme. In particular the invariant distinguishes different Jordan partitions over a fixed multiple discriminant root.*

Proof. The first implication follows by reconstructing R , then choosing an invertible member A and a complementary member B . For completeness, the relevant classical sufficiency can be seen directly. Put $C = A^{-1}B$; it is self-adjoint for A . If $hC = C'h$ identifies the Jordan forms for another pair (A', B') , set

$$H = A^{-1}h^t A'h.$$

Then H is invertible, A -self-adjoint, and commutes with C . There is a polynomial k in H with $k^2 = H$: on each nonzero eigenvalue choose a square root and truncate its Taylor series to the size of the largest Jordan block, then combine by the Chinese remainder theorem. Thus k is invertible, self-adjoint, and commutes with C . The map hk^{-1} is an isometry from A to A' and intertwines C, C' , so it simultaneously identifies A, B with A', B' .

Changing the basis of the pencil replaces C by a fractional linear expression with invertible denominator. At each eigenvalue its derivative is nonzero; hence each Jordan-block size is preserved while its root moves by the same projective transformation. A parameter change can avoid infinity for both finite spectra. Root-labelled elementary divisors therefore suffice, and (4.7) proves the converse for the failure schemes. This argument recalls the classical classification; the new input is recovery from an unmarked abstract failure scheme. $\square$

Example 4.5 (Identical discriminant, distinct intrinsic failure schemes). Choose distinct nonzero $\mu_3, \dots, \mu_n$, and put

$$S_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Consider the symmetric pairs

$$\begin{aligned} A_s &= I_n, & B_s &= \text{diag}(0, 0, \mu_3, \dots, \mu_n), \\ A_j &= S_2 \oplus I_{n-2}, & B_j &= S_2 J_2 \oplus \text{diag}(\mu_3, \dots, \mu_n). \end{aligned}$$

Their determinant binary forms agree up to a nonzero scalar:

$$\det(sA_s + tB_s) = -\det(sA_j + tB_j) = s^2 \prod_{i=3}^n (s + \mu_i t). \quad (4.9)$$

At the unique double root the ranks are respectively $n-2$ and $n-1$, so no simultaneous projective equivalence exists. The two root partitions are $(1, 1)$ and (2) . Their abstract failure schemes are therefore nonisomorphic although both their reduced failure schemes and their discriminant divisors, including multiplicities, agree. This realizes inside the same multiplication construction the classical loss of elementary-divisor information under passage to the determinant.

Corollary 4.6 (Hyperelliptic isomorphism classes). *Fix $g \geq 2$ and $n = 2g + 2$. On the locus of pencils with simple discriminant, $\widehat{D}_R$ gives an injective invariant of the isomorphism classes of smooth hyperelliptic curves of genus g . All these failure schemes have the same reduction; their nonreduced structures distinguish a $(2g-1)$ -dimensional family of curve classes. The pencil invariant also distinguishes the two regular boundary pencils in Example 4.5, where their singular discriminant double covers alone do not.*

Proof. The classical curve associated with a simple-discriminant pencil is the double cover of its parameter line branched at $\det(sA + tB) = 0$; see [41, §4.2, equation (4.3)]. Choose an invertible member. Distinct eigenvalues of $A^{-1}B$ have mutually A -orthogonal eigenlines, each nonisotropic,

so simultaneous diagonalization identifies the pencil with $\langle \sum x_i^2, \sum \lambda_i x_i^2 \rangle$. The unordered branch divisor modulo PGL_2 therefore determines the pencil, and every reduced divisor of degree $2g + 2$ arises. A smooth curve of genus at least two has a unique degree-two map to a rational curve up to automorphism of that curve: two distinct such maps would give a birational map to a curve of bidegree $(2, 2)$ in $\mathbb{P}^1 \times \mathbb{P}^1$, whose arithmetic genus is one. Thus curve isomorphism is exactly branch-divisor equivalence. Applying Theorem 4.3 gives the asserted injectivity and well-definedness.

Ordered distinct branch points have dimension $2g + 2$. Normalizing three of them removes the three-dimensional projective action; taking the finite permutation quotient leaves dimension $2g - 1$. Equality of all reductions follows from (4.4). Finally the two binary forms in (4.9) differ only by a scalar, a square over $\mathbb{C}$, and define isomorphic double covers, whereas their failure schemes have already been separated. $\square$

These are statements about complex isomorphism classes. They do not identify moduli stacks, assert a universal family over a coarse moduli space, or eliminate the different automorphism groups of pencils and curves. Nor is the classical pencil–hyperelliptic correspondence a new Torelli theorem. The conclusion here is reconstruction of those objects and of the elementary-divisor information on their boundary from the abstract nonreduced multiplication-failure scheme, while all reduced failure data are fixed.

5. THE EXACT INFINITESIMAL ORDER OF PENCIL RECONSTRUCTION

The all-pencil theorem admits a finite-order strengthening. It identifies the first infinitesimal neighbourhood that retains the pencil, and shows that every preceding neighbourhood is independent of it. We use ideal-adic neighbourhoods of the deepest stratum; these are not arc spaces or jet schemes.

Throughout this section let $n \geq 3$, $W = \mathrm{Sym}^2 V$, and

$$N = \binom{n+1}{2}, \quad p = N - 2, \quad d = n + 2p = n^2 + 2n - 4. \quad (5.1)$$

For a pencil $R \subset W$, put $S_R = W/R$ and $B = \mathrm{Gr}(n, S_R) \subset \widehat{D}_R$. This notation for the socle Grassmannian is local to this section; the cube-zero algebra is denoted by A_R , as in Section 4. Write $\mathfrak{b} \subset \mathcal{O}_{\widehat{D}_R}$ for the ideal of B , and define

$$Z_R^{[k]} = (B, \mathcal{O}_{\widehat{D}_R} / \mathfrak{b}^{k+1}) \quad (k \geq 0). \quad (5.2)$$

Thus $Z_R^{[0]} = B$. No embedding or identification of its reduction is supplied with the abstract scheme $Z_R^{[k]}$.

Theorem 5.1 (Sharp finite-neighbourhood reconstruction). *For any two pencils $R, R' \subset \mathrm{Sym}^2 V$, the following conditions are equivalent:*

$$\begin{aligned} Z_R^{[d]} \simeq Z_{R'}^{[d]} &\iff \widehat{D}_R \simeq \widehat{D}_{R'} \\ &\iff R' = gR \text{ for some } g \in \mathrm{PGL}(V). \end{aligned} \quad (5.3)$$

In contrast, for every $0 \leq k < d$, all the schemes $Z_R^{[k]}$ are isomorphic. Consequently d is the smallest uniform order at which these neighbourhoods classify all pencils up to projective equivalence. Singular pencils are included.

5.1. The first relation in the conormal algebra. Let $\mathcal{U}$ be tautological on B . The open set of X_R where projection to S_R is injective is canonically $\text{Tot } E$, where $E = \text{Hom}(\mathcal{U}, V)$: a plane is the graph of a unique map from its image in S_R to V . In particular this is one open neighbourhood of all of B , not merely a collection of unrelated formal charts. Let $\mathfrak{a}$ be its zero-section ideal. The exact multiplication calculation (4.4) gives

$$\mathcal{I}_R = (\det T)I_p(\gamma_R \text{Sym}^2 T) \subset \text{Sym } E^*. \quad (5.4)$$

It is generated in degree d , with a nonzero degree- d piece K_R . Indeed each quadratic minor has degree $2p$, and an invertible T makes $\gamma_R \text{Sym}^2 T$ surjective. There are no terms of lower degree, even on special pencils. Homogeneity is an equality of ideals in the actual graph neighbourhood, not just an equality of initial supports.

Lemma 5.2 (Intrinsic recovery from the first relation). *The abstract scheme $Z_R^{[d]}$ determines, as a graded $\mathcal{O}_B$ -algebra, the complete normal-cone algebra $\text{gr}_{\mathfrak{b}} \mathcal{O}_{\widehat{D}_R}$.*

Proof. The reduction of $Z_R^{[d]}$ is B , since its zero-section ideal is nilpotent and B is reduced. Hence its nilradical $\mathfrak{n}$ is intrinsic. Put $L = \mathfrak{n}/\mathfrak{n}^2$. Formula (5.4) and $d > 1$ identify $L = E^*$. Multiplication in the given finite scheme recovers

$$K_R = \ker[\text{Sym}^d L \longrightarrow \mathfrak{n}^d/\mathfrak{n}^{d+1}]. \quad (5.5)$$

The last denominator is zero, but retaining it displays the grading. No splitting of the reduction map is needed to form this kernel. Because $\mathcal{I}_R$ is generated in this single degree,

$$\text{gr}_{\mathfrak{b}} \mathcal{O}_{\widehat{D}_R} = \text{Sym}_{\mathcal{O}_B} L/(K_R). \quad (5.6)$$

Here (K_R) denotes the homogeneous ideal generated by K_R , not its radical. The right side is intrinsic to the finite scheme, proving the assertion. $\square$

Proof of Theorem 5.1. For $k < d$, equations (5.4) and (5.2) give

$$\mathcal{O}_{Z_R^{[k]}} = \text{Sym } E^* / \mathfrak{a}^{k+1}.$$

A linear identification $S_R \simeq S_{R'}$ identifies B , its tautological bundle, and E . It therefore identifies every one of these lower-order neighbourhoods.

An abstract isomorphism at order d induces an isomorphism of reductions, of conormal bundles, and, by Lemma 5.2, of complete graded normal cones. We spell out why the reconstruction in Theorem 4.1 requires no information outside this cone. Its reduced support is the square-matrix determinant cone; its iterated singular loci give the rank-one cone. The relative scheme of maximal linear spaces in that cone has the two rulings $\mathbb{P}(V) \times B$ and $\mathbb{P}_B(\mathcal{U}^*)$. Exactly the first is projectively trivial, as is seen by restriction to a Schubert line. Thus it is intrinsically oriented. The induced projective transformation of the trivial factor is constant on the connected projective base. Local lifts have the form $T \mapsto gTh^{-1}$ for this one $[g]$. The determinant ideal line is intrinsic, so its ideal quotient in (5.4) recovers the residual ideal. Finally Lemma 4.2 identifies the entire left coefficient line $\mathbb{C}\beta_R$, and exterior duality (4.8) recovers the Pluecker line of R . These are operations on the recovered normal cone; we do not assume that its isomorphism extends to X_R or $\widehat{D}_R$. They imply $R' = gR$.

Conversely such a projective transformation induces the algebra and failure-scheme isomorphisms, carrying the socle Grassmannian to the socle Grassmannian and preserving all its ideal powers. This proves (5.3), also using Theorem 4.3. For sharpness, there are inequivalent pencils in every dimension under consideration: $\langle x_1^2, x_1x_2 \rangle$ has a common linear factor, whereas $\langle x_1^2, x_2^2 \rangle$ does not. Their neighbourhoods are nevertheless isomorphic at every order below d . Thus no smaller order classifies all pencils. $\square$

5.2. A closed immersion of the full parameter scheme. The finite-order result has a scheme-theoretic parameter version, which records infinitesimal variation as well as closed points. Here tensor coordinates are fixed. This additional qualification is important: we are not asserting an equivalence of unmarked moduli stacks.

Fix an auxiliary n -dimensional vector space U , and put

$$F(V) = \bigwedge^p \mathrm{Sym}^2 V, \quad F(U) = \bigwedge^p \mathrm{Sym}^2 U, \quad M = \dim F(U) = \binom{N}{2}, \quad H = \mathrm{Hom}(U, V).$$

Choose a nonzero scalar representative δ of the determinant polynomial on H ; its scalar has no effect on the subspaces below. Let $K_R \subset \mathrm{Sym}^d H^*$ be the degree- d space of $(\det T)I_p(\gamma_R \mathrm{Sym}^2 T)$ in these fixed coordinates.

Theorem 5.3 (Faithful finite-order parameterization). *The assignment*

$$\Phi : \mathrm{Gr}(2, \mathrm{Sym}^2 V) \longrightarrow \mathrm{Gr}(M, \mathrm{Sym}^d H^*), \quad R \longmapsto K_R, \quad (5.7)$$

is a closed immersion. It is defined on the whole Grassmannian, including singular pencils. Its construction and its inverse on the image commute with arbitrary base change over $\mathbb{C}$. In particular it separates families over nonreduced bases and has injective differential at every pencil.

Lemma 5.4 (Tensoring a line with a fixed space). *For finite-dimensional vector spaces A, B_0 , with $b = \dim B_0 > 0$, the map $\mathbb{P}(A) \rightarrow \mathrm{Gr}(b, A \otimes B_0)$, $\ell \mapsto \ell \otimes B_0$, is a closed immersion.*

Proof. Choose bases $a_0, \dots, a_t$ and a basis of B_0 . Above the projective chart $\ell = \mathbb{C}(a_0 + \sum x_i a_i)$, the subspace is the graph of the block matrices $x_i I_b$. The corresponding Grassmannian chart has arbitrary $b \times b$ blocks. The image is cut out by their off-diagonal entries and the differences between their diagonal entries; an inverse reads one diagonal entry in each block. These are scheme-theoretic equations over any $\mathbb{C}$ -algebra, not just pointwise conditions. The preimage of this Grassmannian chart is exactly the stated projective chart, since the projection has determinant a_0^b . Such charts cover the image. The map is proper because its source is projective, so these chartwise closed immersions give a global closed immersion. $\square$

Proof of Theorem 5.3. The coefficient map (4.6) becomes

$$\mu : F(V)^* \otimes F(U) \longrightarrow \mathrm{Sym}^{2p} H^*.$$

It is injective: by Lemma 4.2 its source is irreducible for $\mathrm{GL}(V) \times \mathrm{GL}(U)$, and evaluation at an isomorphism $U \simeq V$ shows that the equivariant map is nonzero. Multiplication by the nonzero polynomial δ is injective. Thus $\nu = \delta\mu$ is a fixed linear inclusion and

$$K_R = \nu(\mathbb{C}\beta_R \otimes F(U)). \quad (5.8)$$

In particular $\dim K_R = M$, also on every singular pencil. By exterior duality the quotient-volume map $R \mapsto \mathbb{C}\beta_R$ is the Pluecker closed immersion $\mathrm{Gr}(2, W) \hookrightarrow \mathbb{P}(F(V)^*)$. Factor Φ through this map, the closed immersion of Lemma 5.4, and the Grassmannian closed immersion induced by ν . This proves the theorem. All are morphisms of schemes defined by the displayed polynomial and linear equations. A closed immersion is a monomorphism and remains one after arbitrary base change. This proves the family assertion, not merely separation of complex-valued points. $\square$

Corollary 5.5 (A flat family of finite normal algebras). *Over $\mathrm{Gr}(2, W)$, the algebras*

$$Q_R = \mathrm{Sym} H^* / ((K_R) + \mathfrak{m}^{d+1}), \quad \mathfrak{m} = \bigoplus_{j>0} \mathrm{Sym}^j H^*, \quad (5.9)$$

form a finite locally free family of rank

$$\binom{n^2 + d}{d} - \binom{N}{2}.$$

In fixed coordinates this family determines every base-changed family of pencils. All its truncations below degree d are constant families.

Proof. Theorem 5.3 makes the spaces K_R a rank- M subbundle of the constant degree- d bundle, with locally free quotient. In degree $j < d$ the algebra is $\text{Sym}^j H^*$, and its degree- d piece is $\text{Sym}^d H^*/K_R$; higher degrees vanish. This proves local freeness and base-change compatibility. Summing the ranks gives the displayed binomial identity. The kernel of the coordinate quotient in degree d recovers K_R , and Φ recovers the pencil family. This is a statement about coordinate quotient algebras; it does not discard their coordinate maps and then infer a family reconstruction theorem. $\square$

5.3. Classical moduli at a sharp order.

Corollary 5.6 (Hyperelliptic classes in finite neighbourhoods). *Let $g \geq 2$, $n = 2g + 2$, and $d = 4g^2 + 12g + 4$. For pencils with simple discriminant, $Z_R^{[d]}$ and $Z_{R'}^{[d]}$ are isomorphic if and only if their associated hyperelliptic curves are isomorphic. Every lower-order neighbourhood is independent of the curve. Thus the same $(2g - 1)$ -dimensional family of curve classes is invisible to all these lower-order neighbourhoods and is separated at order d .*

Proof. Combine Theorem 5.1 with Corollary 4.6, and substitute $n = 2g + 2$ in (5.1). $\square$

The order here concerns a neighbourhood of the entire projective socle Grassmannian. Its global ruling distinguishes the constant V -factor from the tautological factor. A single unmarked Artinian normal slice does not supply this orientation; the coordinate statement of Corollary 5.5 is deliberately different. The result is therefore not a claim that one unmarked local algebra replaces the projective family. The matrix-pencil classification and the hyperelliptic correspondence remain classical. The assertion proved here is that this classical information first appears at the specified order in the intrinsic multiplication-failure neighbourhoods.

6. SPECTRAL TORSION AND RANK-PRESERVING SPECIALIZATION

The finite-neighbourhood theorem recovers more than the discriminant or the corank of each member of a regular pencil. We identify its classical spectral invariant and then give a flat specialization on which the entire discriminant and every reduced rank locus remain constant, while the recovered invariant changes. The two specializations are separated at exactly the order of Theorem 5.1. The parameter space throughout is the usual Grassmannian of pencils; the failure map is the original multiplication map, not an added coefficient-evaluation block.

6.1. The spectral torsion sheaf. Let $R \subset \text{Sym}^2 V$ be a regular pencil, meaning that it contains a nonsingular quadratic form. On its parameter line $L_R = \mathbb{P}(R)$, the tautological quadratic form gives an exact sequence

$$0 \longrightarrow V^* \otimes \mathcal{O}_{L_R}(-1) \xrightarrow{q_R} V \otimes \mathcal{O}_{L_R} \longrightarrow \mathcal{T}_R \longrightarrow 0. \quad (6.1)$$

The first arrow is injective because its determinant is nonzero at the generic point of the smooth integral curve L_R . Thus $\mathcal{T}_R$ is a torsion sheaf of length n , whose zeroth Fitting divisor is the full discriminant. The pair $(L_R, \mathcal{T}_R)$ is considered up to isomorphism of the parameter line. Its definition involves no choice of a basis of R . The relation with elementary divisors is classical; see [40, §2, Corollary 2.1 and equation (5)]. We record the argument to make precise which datum is recovered.

Lemma 6.1 (Sheaves on the affine spectral line). *Let A, B be symmetric complex matrices with A invertible. On the affine line with coordinate z , the cokernel of $zA - B$ is the finite-dimensional $\mathbb{C}[z]$ -module on $\mathbb{C}^n$ for which multiplication by z is $C = A^{-1}B$. An isomorphism of such sheaves, in this fixed coordinate, is equivalent to similarity of the corresponding matrices.*

Proof. Multiplication on the target by A^{-1} identifies the presentation with $zI - C$. Reduction of a polynomial vector modulo these relations sends $\sum_j z^j v_j$ to $\sum_j C^j v_j$; constant vectors give the inverse identification. The equivalence between coherent sheaves on an affine scheme and finitely generated modules identifies a finite-length sheaf supported in this affine line with this module. The support is finite, so its underlying complex vector space is finite dimensional. A module isomorphism is a linear isomorphism G satisfying $GC = C'G$, which is exactly similarity. Extension by zero across a point outside the support loses no sheaf data. $\square$

Lemma 6.2 (Polynomial square roots of invertible operators). *For every invertible complex matrix K there is $h \in \mathbb{C}[x]$ with $h(K)^2 = K$. If K is self-adjoint for a nondegenerate symmetric form and commutes with C , then $h(K)$ has both properties.*

Proof. Write the minimal polynomial as $\prod_\lambda (x - \lambda)^{a_\lambda}$, where $\lambda \neq 0$. In its local Artinian factor choose a square root $\mu_\lambda^2 = \lambda$, and put

$$h_\lambda(x) = \mu_\lambda \sum_{j=0}^{a_\lambda-1} \binom{1/2}{j} \left(\frac{x - \lambda}{\lambda} \right)^j.$$

The formal binomial identity, truncated modulo $(x - \lambda)^{a_\lambda}$, gives $h_\lambda(x)^2 = x$. The Chinese remainder theorem joins these polynomials to a single class modulo the minimal polynomial. Evaluation at K proves the assertion even when K is not semisimple. Polynomials in one self-adjoint operator remain self-adjoint, and every polynomial in K commutes with every operator commuting with K . $\square$

Theorem 6.3 (Spectral reconstruction at the first relation). *Let $n \geq 3$, $d = n^2 + 2n - 4$, and let R, R' be regular pencils. The following are equivalent:*

- (1) *the unmarked schemes $Z_R^{[d]}$ and $Z_{R'}^{[d]}$ are isomorphic;*
- (2) *there is an isomorphism $f : L_R \rightarrow L_{R'}$ with $\mathcal{T}_R \simeq f^* \mathcal{T}_{R'}$;*
- (3) *the pencils R, R' are projectively equivalent.*

In particular the finite scheme determines the full root-labelled spectral torsion sheaf, not only its Fitting divisor or its support. This is an equivalence of isomorphism classes of objects; it is not an assertion identifying their automorphism groups or moduli stacks.

Proof. The equivalence of the first and third conditions is Theorem 5.1. A congruence of pencils induces an isomorphism of (6.1), so the third implies the second.

For the converse choose a common affine coordinate whose point at infinity avoids the two spectral supports, using f to identify the lines. Write the corresponding symmetric pairs as (A, B) and (A', B') , with A, A' invertible, and set $C = A^{-1}B$, $C' = A'^{-1}B'$. Choose the presentation $zA - B$ for the affine coordinate on both lines. Lemma 6.1 identifies the two sheaves with the modules of C, C' , so their isomorphism gives a similarity. After applying that change of coordinates, assume $C' = C$. Both A and A' are nondegenerate symmetric forms for which C is self-adjoint. Then $K = A^{-1}A'$ commutes with C and is self-adjoint for A . Lemma 6.2 provides a polynomial h with $h(K)^2 = K$. Put $H = h(K)$. It commutes with C , is self-adjoint for A , and satisfies

$$H^t A H = A', \quad H^t B H = H^t A C H = A' C = B'.$$

Thus the pairs, and hence the pencils, are congruent. This also shows why the spectral sheaf alone suffices here; no additional pairing on that sheaf is required over $\mathbb{C}$. $\square$

For $x \in \text{Supp } \mathcal{T}_R$, put

$$\ell_a(x) = \text{length}(\mathcal{T}_{R,x} / \mathfrak{m}_x^a \mathcal{T}_{R,x}), \quad a \geq 0. \tag{6.2}$$

If the elementary-divisor exponents at x are $e_1, \dots, e_b$, the structure theorem over the discrete valuation ring $\mathcal{O}_{L_R, x}$ gives

$$\ell_a(x) = \sum_{j=1}^b \min(a, e_j), \quad \ell_a(x) - \ell_{a-1}(x) = \#\{j : e_j \geq a\}. \quad (6.3)$$

Thus these intrinsic local lengths recover every exponent, including repeated roots. The corank of the pencil at x gives only $\ell_1(x)$, and the multiplicity of the discriminant gives only the eventual value $\ell_a(x) = \sum_j e_j$. The intermediate values are genuinely additional information.

6.2. A specialization with constant discriminant and coranks. The following family keeps both ends of this length profile fixed. Fix $i \in \mathbb{C}$ with $i^2 = -1$, and set

$$N_0 = \begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}, \quad P_t = \frac{t}{2} N_0, \quad A_0 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \quad C_t = \begin{pmatrix} P_t & I_2 \\ 0 & P_t \end{pmatrix}. \quad (6.4)$$

Put

$$B_t = A_0 C_t = \begin{pmatrix} 0 & P_t \\ P_t & I_2 \end{pmatrix}, \quad R_t = \langle A_0, B_t \rangle \subset \text{Sym}^2 \mathbb{C}^4. \quad (6.5)$$

Both matrices are symmetric and A_0 is invertible. They are independent for every t . For $n > 4$, append I_{n-4} to A_0 and append $\text{diag}(\mu_5, \dots, \mu_n)$ to B_t , where the μ_j are fixed distinct nonzero numbers. Denote the resulting matrices by $A^{(n)}, B_t^{(n)}$ and their pencil again by R_t .

Theorem 6.4 (Separation inside fixed reduced rank data). *For every $n \geq 4$, the family R_t , $t \in \mathbb{A}^1$, has the following properties.*

- (1) *Its full discriminant, with multiplicities, is independent of t :*

$$\det(sA^{(n)} + uB_t^{(n)}) = s^4 \prod_{j=5}^n (s + \mu_j u). \quad (6.6)$$

For every minor size, the reduced determinantal locus on the marked parameter line is also independent of t .

- (2) *At the root $s = 0$, the spectral module has exponents $(3, 1)$ for $t \neq 0$ and $(2, 2)$ for $t = 0$. Consequently $Z_{R_t}^{[d]} \not\cong Z_{R_0}^{[d]}$ for every $t \neq 0$, whereas all the punctured fibres are isomorphic.*
- (3) *The schemes $Z_{R_t}^{[d]}$ form a flat projective family over $\mathbb{A}^1$, with fixed reduction. After one identification of the socle quotient bundles, every truncation of order $k < d$ is a constant family. Thus the first separating order remains exactly $d = n^2 + 2n - 4$ even under the constraints in part (1).*

Proof. The identities $N_0^2 = 0$ and $\text{rank } N_0 = 1$ give

$$C_t^2 = \begin{pmatrix} 0 & tN_0 \\ 0 & 0 \end{pmatrix}, \quad C_t^3 = 0, \quad C_t = \begin{pmatrix} I_2 \\ P_t \end{pmatrix} \begin{pmatrix} P_t & I_2 \end{pmatrix}. \quad (6.7)$$

The right factor in the last expression is surjective and the left factor is injective. Hence $\text{rank } C_t = 2$ for every t , whereas $\text{rank } C_t^2 = 1$ precisely when $t \neq 0$. The nilpotent partitions are therefore $(3, 1)$ and $(2, 2)$. Also $\det A_0 = 1$ and $\det(sI_4 + uC_t) = s^4$, proving (6.6). At $s = 0$ the full pencil has corank two for all t ; at each remaining root it has corank one, and elsewhere corank zero. Equality of these functions on $\mathbb{P}^1(\mathbb{C})$ implies equality of the reduced closed determinantal subschemes for each minor size.

At the unique multiplicity-four root, formula (6.3) yields respectively

$$(\ell_1, \ell_2, \ell_3, \dots) = (2, 3, 4, \dots), \quad (2, 4, 4, \dots).$$

No automorphism of the parameter line can exchange this root with a simple one. Theorem 6.3 separates the special fibre from the punctured fibres and identifies the latter with one another. This is a statement about geometric fibres; no choice of a simultaneous trivializing congruence on $\mathbb{A}^1 \setminus 0$ is needed.

For flatness we use the relative gluing of Theorem 7.1; the following coordinates identify its pullback in the present family. First note that $t \mapsto R_t$ is a morphism to $\mathrm{Gr}(2, \mathrm{Sym}^2 V)$, not merely a collection of pencils. In symmetric matrix coordinates the two fixed linear functionals

$$a(Q) = Q_{13} + iQ_{14}, \quad b(Q) = Q_{33}$$

satisfy $a(A^{(n)}) = b(B_t^{(n)}) = 1$ and $a(B_t^{(n)}) = b(A^{(n)}) = 0$. Their common kernel is a fixed complement to every R_t . It trivializes $S_{R_t} = \mathrm{Sym}^2 V / R_t$, and hence identifies the relative socle Grassmannian with $B \times \mathbb{A}^1$.

On a frame chart of its tautological bundle, the graph description (5.4) identifies the truncated algebra with

$$\bigoplus_{j=0}^{d-1} \mathrm{Sym}^j E^* \oplus (\mathrm{Sym}^d E^* / K_{R_t}). \quad (6.8)$$

By Theorem 5.3, K_{R_t} is a subbundle of constant rank $\binom{N}{2}$, with locally free quotient. This statement commutes with pullback to the present base and with right changes of frame, so the local descriptions glue. Thus (6.8) is locally free on $B \times \mathbb{A}^1$, of rank $\binom{n^2+d}{d} - \binom{N}{2}$. The graph projection is a finite flat morphism to that base. Since B is projective, its composition with projection to $\mathbb{A}^1$ is flat and projective. Below degree d there is no relation, and the fixed complement identifies the entire lower truncated algebra, including multiplication, with a constant family. These arguments prove flatness of the finite neighbourhoods; they make no assertion about flatness of the unrestricted failure schemes. $\square$

The qualifier “reduced” in part (1) is necessary. At $s = 0$, in a local parameter $z = s/u$, the gcd of the three-row minors of the four-dimensional block is a unit times z for $t \neq 0$ and a unit times z^2 for $t = 0$. Both radicals are (z) . Thus the family retains exactly the higher Fitting information discarded by reduction; it does not contradict the classical classification by the full determinantal ideals.

6.3. An independently visible change of reciprocal geometry. For a regular pencil let its reciprocal curve be the projective closure of the inverses of its nonsingular members. These curves and their strata are studied independently of multiplication-failure schemes in [40, §3].

Corollary 6.5 (A conic specializes to a line). *For the four-dimensional family (6.5), the reciprocal curve is a smooth plane conic when $t \neq 0$ and a projective line when $t = 0$. The unmarked twentieth neighbourhood distinguishes these two cases, while all earlier neighbourhoods, the full discriminant, and all reduced rank loci agree.*

Proof. For $s \neq 0$, multiplication on the right by the fixed invertible matrix A_0^{-1} identifies the projective inverse with

$$(sI_4 + uC_t)^{-1} = s^{-1}I_4 - us^{-2}C_t + u^2s^{-3}C_t^2.$$

If $t \neq 0$, the matrices I_4, C_t, C_t^2 are linearly independent: their span has dimension equal to the degree three of the minimal polynomial. The closure is therefore the quadratic Veronese image $[s : u] \mapsto [s^2 : -su : u^2]$, a smooth conic in its spanning plane. At $t = 0$ the last term vanishes, and I_4, C_0 are independent, giving a line. The remaining assertions follow from Theorem 6.4, with $d = 20$. The degree drop does not assert that the reciprocal curves themselves form a flat family; the flat family proved above consists of the finite failure neighbourhoods. $\square$

7. THE UNIVERSAL FINITE NEIGHBOURHOOD AND SPECTRAL STRATA

The finite-order construction is a family of actual neighbourhoods in multiplication-failure schemes. This fact is needed when passing from individual pencils to degenerations and infinitesimal parameter spaces. We establish it before discussing the spectral incidence schemes.

Put $P = \text{Gr}(2, \text{Sym}^2 V)$, and write $\mathcal{R}$ and $\mathcal{S} = (\text{Sym}^2 V \otimes \mathcal{O}_P)/\mathcal{R}$ for its universal relation and quotient bundles. Let

$$\pi : \mathcal{B} = \text{Gr}_P(n, \mathcal{S}) \longrightarrow P, \quad \mathcal{E} = \text{Hom}(\mathcal{U}, V \otimes \mathcal{O}_{\mathcal{B}}),$$

where $\mathcal{U}$ is tautological of rank n . As before, $N = n(n+1)/2$, $p = N - 2$, $d = n + 2p$, and $M = \binom{N}{2}$.

Theorem 7.1 (Universal neighbourhood and arbitrary base change). *The universal multiplication-failure scheme has a canonical order- d neighbourhood $\mathcal{Z}^{[d]}$ of $\mathcal{B}$. There is a rank- M vector subbundle $\mathcal{K} \subset \text{Sym}^d \mathcal{E}^*$ such that*

$$\mathcal{Z}^{[d]} = \text{Spec}_{\mathcal{B}} \left(\text{Sym} \mathcal{E}^* / (\mathcal{K}, \text{Sym}^{d+1} \mathcal{E}^*) \right). \quad (7.1)$$

This is finite locally free over $\mathcal{B}$, of rank

$$\binom{n^2 + d}{d} - \binom{N}{2}, \quad (7.2)$$

and is projective and flat over P . Its construction, its ideal-adic filtration, and its identification with the actual multiplication-failure neighbourhood commute with every morphism of complex schemes $T \rightarrow P$, including nonreduced ones. For $k < d$, its order- k truncation is $\text{Spec}_{\mathcal{B}} \bigoplus_{j=0}^k \text{Sym}^j \mathcal{E}^$, with truncated multiplication. After trivializing $\mathcal{S}$ on the parameter base, these lower-order families are constant.*

Proof. The locus in $\text{Gr}_P(n, V \oplus \mathcal{S})$ on which projection to $\mathcal{S}$ is injective is canonically $\text{Tot } \mathcal{E}$. It is an open neighbourhood of $\mathcal{B}$, not a choice of formal coordinates. On a frame chart of $\mathcal{U}$, let $T : \mathcal{U} \rightarrow V$ be the universal graph map and let $\gamma : \text{Sym}^2 V \rightarrow \mathcal{S}$ be the quotient. The same block elimination that proves (5.4) works over its coordinate ring: the ideal of the multiplication-failure scheme is

$$(\det T) I_p(\gamma \text{Sym}^2 T).$$

Every generator is homogeneous of degree d .

Here is also a coordinate-free description of its first relation bundle. The coefficient map, followed by determinant multiplication, is a morphism

$$\det V^* \otimes \det \mathcal{U} \otimes \det \mathcal{S}^* \otimes \bigwedge^p \text{Sym}^2 \mathcal{U} \longrightarrow \text{Sym}^d \mathcal{E}^*. \quad (7.3)$$

The inclusion used in this expression is $\det \mathcal{S}^* \hookrightarrow (\bigwedge^p \text{Sym}^2 V)^*$. The determinant factors evaluate $\det T$, and the remaining factors evaluate $\bigwedge^p \text{Sym}^2 T$ followed by the quotient. Thus the image is exactly the span of the displayed maximal minors, not a separately inserted coefficient module.

On a frame chart this map is the fixed injective linear coefficient map ν of Theorem 5.3, restricted to a line subbundle tensored with the fixed space $\bigwedge^p \text{Sym}^2 \mathbb{C}^n$. Its image is a subbundle of rank M : locally split the line in the source, and split the fixed complex linear inclusion ν . Both splittings remain splittings over any complex algebra. Denote its image by $\mathcal{K}$. A right change of frame of $\mathcal{U}$ changes the graph coordinates and this coefficient map contragrediently. The determinant character in (7.3) is exactly the one arising from $\det T$. Hence these local ideals and their degree- d subbundles agree on overlaps. In particular, they glue as ideals of the given graph neighbourhood.

Let $\mathfrak{b}$ be the ideal of the zero section inside the multiplication-failure scheme. In graph coordinates its preimage is the positive-degree ideal. Quotienting by $\mathfrak{b}^{d+1}$ therefore gives precisely (7.1). This identifies the gluing with the actual ideal-adic neighbourhood, not merely with algebras having the correct geometric fibres. Its underlying module is

$$\bigoplus_{j=0}^{d-1} \mathrm{Sym}^j \mathcal{E}^* \oplus (\mathrm{Sym}^d \mathcal{E}^* / \mathcal{K}).$$

All terms are locally free. Counting monomials in n^2 variables gives (7.2). The morphism $\mathcal{B} \rightarrow P$ is smooth and projective, so the asserted flatness and projectivity follow. Finally the graph description, the split coefficient inclusion, and the homogeneous quotient all commute with base change. Below degree d the ideal contributes no relation. This proves the final assertion as well. $\square$

Remark 7.2. Over a nonreduced base, the theorem concerns the specified relative socle subscheme $\mathcal{B}_T$ and its ideal. It does not identify that ideal with the absolute nilradical of $\mathcal{Z}_T^{[d]}$, which can also contain nilpotents from the base. On geometric fibres $\mathcal{B}_t$ is exactly the reduction. Nor is this theorem an equivalence of unmarked deformation groupoids: extra automorphisms of truncated algebras are not being identified with pencil congruences.

7.1. Spectral incidence schemes from the first relation space. On the regular-pencil locus, pass locally to ordered generators (A, B) with A invertible, and put $C = A^{-1}B$. For a marked point with affine coordinate z , set $X = C - zI$. The next elementary presentation works over an arbitrary coefficient ring; no diagonalization or division by eigenvalue differences is involved.

Lemma 7.3 (Truncated spectral presentation). *Let D be a commutative ring, $X \in \mathrm{Mat}_n(D)$, and $a \geq 1$. There is a natural isomorphism*

$$\mathrm{coker}(wI - X) \otimes_{D[w]} D[w]/(w^a) \simeq \mathrm{coker}(X^a : D^n \rightarrow D^n). \quad (7.4)$$

Equivalently, the a -square block presentation with diagonal blocks X and subdiagonal blocks $-I$ is equivalent, by invertible row and column operations over D , to $I_{n(a-1)} \oplus X^a$.

Proof. Map the class of $\sum_{j=0}^{a-1} w^j v_j$ to the class of $\sum_{j=0}^{a-1} X^j v_j$ modulo $X^a D^n$. The relation $wv = Xv$ and the relation $w^a v = 0$ show that this map is well-defined. Sending a constant vector to its class is inverse. Eliminating successively the blocks with coefficient $-I$ gives the stated presentation equivalence, using only unit pivots. This also proves naturality under ring homomorphisms. $\square$

Theorem 7.4 (Scheme-theoretic spectral readout). *In the fixed tensor presentation of Theorem 5.3, the first relation subspace determines, on the regular-pencil locus, every closed spectral incidence scheme*

$$\mathfrak{S}_{a,h} = V(I_{n-h+1}((C - zI)^a)), \quad 1 \leq a, h \leq n. \quad (7.5)$$

This determination commutes with arbitrary complex base change. At a geometric point, membership is equivalent to $\mathrm{length}(\mathcal{T}_z / \mathfrak{m}_z^a \mathcal{T}_z) \geq h$. Consequently the schemes of specified spectral length profiles, and not only their reduced supports, can be read from the order- d relation subspace.

Proof. The closed immersion Φ of Theorem 5.3 has an inverse on its scheme-theoretic image. Thus a family of relation subspaces in that image recovers its morphism to P , over any base, and hence its universal pencil. In the displayed chart its spectral sheaf has presentation $(z+w)I - C = wI - X$. Lemma 7.3 gives the presentation X^a for restriction to the order- a fat point centred at z . Its $(h-1)$ -st Fitting ideal is exactly the ideal in (7.5); Fitting ideals of finitely presented modules commute with base change. On a field the cokernel has dimension at least h exactly when these minors vanish.

These descriptions are independent of the chosen frames. A change of basis in V conjugates X ; a change of affine coordinate replaces the parameter ideal at the marked point by a unit multiple locally. Alternatively, restriction of the spectral sheaf to the fat diagonal on its parameter line defines the same finite module without a coordinate. Thus the Fitting ideals glue. Rank equality is imposed by the corresponding rank inequality together with the open complement of the next inequality. Intersecting these locally closed conditions gives each specified length profile. Formula (6.3) then identifies its geometric fibres with the prescribed elementary-divisor partition. $\square$

7.2. Realizing the complete Segre stratification. Let P_σ denote the reduced closure in P of a regular Segre stratum, and let $J_\sigma = \Phi(P_\sigma)$. Because Φ is a closed immersion, it gives isomorphisms

$$P_\sigma \simeq J_\sigma, \quad P_{\sigma_1} \cap_P \cdots \cap_P P_{\sigma_s} \simeq J_{\sigma_1} \cap_{\Phi(P)} \cdots \cap_{\Phi(P)} J_{\sigma_s}, \quad (7.6)$$

where the intersections carry their scheme structures. In particular their local rings and formal completions are preserved. Pulling back Theorem 7.1 realizes these parameter schemes by flat projective families of the actual finite failure neighbourhoods. The classical inclusions among Segre-stratum closures [40, Theorem 5.1] are therefore all realized, rather than just the one degeneration of Theorem 6.4.

There are two distinct assertions here. The Segre closure order is classical, and transport of closed subschemes through a closed immersion is formal. The geometric input proved here is the base-change-compatible realization by actual multiplication-failure neighbourhoods and the recovery of their spectral Fitting schemes. The ambient parameter space in (7.6) is the image of Φ , not a moduli space of every possible unmarked deformation of a finite scheme.

8. FIXED-DISCRIMINANT STRATA AND ALL ADMISSIBLE SPECIALIZATIONS

The universal family of Section 7 and the example of Section 6 lead to a classification of the entire fixed-discriminant, fixed-reduced-rank locus. The spectral orbit order is classical. Every allowed specialization is realized here by a flat family of the original finite multiplication-failure neighbourhoods, and the same first relation detects every strict transition.

Fix distinct $\alpha_1, \dots, \alpha_s \in \mathbb{C}$, positive m_i with $\sum m_i = n \geq 3$, and $1 \leq c_i \leq m_i$. Fix a nonsingular symmetric A , and let $\mathfrak{p}_A$ be its self-adjoint endomorphisms. Consider the reduced locally closed variety

$$X_{\mathbf{m}, \mathbf{c}}(A) = \left\{ C \in \mathfrak{p}_A : \det(zI - C) = \prod_i (z - \alpha_i)^{m_i}, \quad \dim \ker(C - \alpha_i I) = c_i \right\}. \quad (8.1)$$

Exclude $s = 1, c_1 = n$, for which C is scalar and $\langle A, AC \rangle$ is not a pencil. Every other datum is realized. Put $R_C = \langle A, AC \rangle$, $d = n^2 + 2n - 4$, and $Z_C = Z_{R_C}^{[d]}$. The full binary determinant is $\det(A) \prod_i (u + \alpha_i v)^{m_i}$. The coranks and labelled roots specify all reduced determinantal loci on the line.

Write $\mathcal{P}(m, c)$ for partitions into exactly c positive parts. For equal-size partitions, $\mu \preceq \lambda$ means $\sum_{j \leq a} \mu_j \leq \sum_{j \leq a} \lambda_j$ for every a ; write λ' for the conjugate partition.

Theorem 8.1 (Classification and specialization at the first relation). *The following statements hold.*

- (1) *The $O(A)$ -orbits in $X_{\mathbf{m}, \mathbf{c}}(A)$ are indexed by $\lambda \in \prod_i \mathcal{P}(m_i, c_i)$, with dimensions*

$$\dim \mathcal{O}_\lambda = \frac{1}{2} \left(n^2 - \sum_i \sum_a (\lambda'_{i,a})^2 \right). \quad (8.2)$$

Unmarked isomorphism classes of Z_C are these tuples modulo the permutation group induced on the roots by projective transformations preserving their multiplicities and coranks.

- (2) A specialization from λ to μ inside (8.1) exists if and only if

$$\mu_i \preceq \lambda_i \quad \text{for every } i. \quad (8.3)$$

Every such pair is realized over a smooth pointed algebraic curve, after shrinking at the special point, with A fixed and with constant full determinant and reduced rank loci.

- (3) Every such pencil family induces a flat projective family of Z_C . After shrinking the curve, every truncation of order $k < d$ is constant. If at least one dominance inequality is strict, the generic and special Z_C are nonisomorphic; their first distinguishing order is d .
- (4) The recovered spectral module at α_i has length profile

$$L_{i,a} = \sum_j \min(a, \lambda_{i,j}). \quad (8.4)$$

Every $L_{i,a}$ can only increase in the stated specialization. All these inequalities together are equivalent to (8.3), giving an intrinsic numerical criterion for every admissible transition.

Proof. We specify the classical input. On a complex nondegenerate symmetric space of dimension m , full orthogonal orbits of self-adjoint nilpotents are indexed by all partitions λ of m . Their dimensions are $(m^2 - \sum_a (\lambda'_a)^2)/2$, and the closure order is dominance. These are the statements in [44, §1, p. 3, equation (2) and the following dominance criterion]. They concern self-adjoint matrices, not the skew-adjoint orthogonal Lie algebra. The Grassmannian Segre-stratum comparison is [40, Theorem 5.1].

The primary spaces $V_i = \ker(C - \alpha_i I)^{m_i}$ are an orthogonal direct sum: the Chinese remainder idempotents are polynomials in the self-adjoint C , hence self-adjoint projections. Each restricted form is nondegenerate, and $C|_{V_i} = \alpha_i I + N_i$, with N_i self-adjoint nilpotent. The prescribed corank is its number of Jordan blocks. The classical theorem gives the indexing. Conversely, a Jordan block is self-adjoint for the reversed-identity Gram matrix; an isometry with the chosen nondegenerate form realizes each partition.

The space of orthogonal decompositions has dimension $(n^2 - \sum_i m_i^2)/2$. Adding the nilpotent orbit dimensions proves (8.2). These are dimensions in the fixed- A matrix space, not in an unmarked moduli stack. Full orthogonal orbits need not be connected; no irreducibility assertion is needed. Theorem 6.3 identifies the unmarked neighbourhood with the pencil, and its spectral module determines the partition at each root. A projective change of the line can therefore do exactly the stated root permutations. For one or two roots the projective stabilizer may be infinite; only its finite permutation image is used.

For necessity, the same polynomial idempotents decompose a family with fixed characteristic polynomial into subbundles of ranks m_i . The rank of N_i^a cannot increase at specialization. Since $m_i - \text{rank } N_i^a = \sum_j \min(a, \lambda_{i,j})$, this gives the length inequalities. Summing the first a columns of conjugate Young diagrams shows that they are equivalent to dominance: conjugation reverses the dominance order.

For sufficiency, fix an orthogonal decomposition of (V, A) . The closure criterion puts a chosen representative of μ_i in the closure of the orbit for λ_i . Thus the special tuple lies in the closure of the product of the generic orbits. Choose an irreducible component of that closure containing the point and meeting the generic orbit product. Successive general hyperplane sections through the point, avoiding the closed complement, give an integral algebraic curve with generic point in that open orbit product. Normalize it and choose a point over the prescribed point. A normal complex curve is smooth. Remove the other points outside the generic orbit product. A constant curve suffices in the equality case. The resulting block operator $\bigoplus_i (\alpha_i I + N_i(t))$ is self-adjoint for

the same A . Its determinant is unchanged, and its coranks agree generically and specially, hence throughout the shrunk curve. This gives the required specialization. It does not assert constancy of the full higher Fitting ideals.

For part (3), pull back Theorem 7.1 along the curve's morphism to $P = \text{Gr}(2, \text{Sym}^2 V)$. That theorem identifies the family with the actual ideal-adic neighbourhoods and proves flatness and projectivity under arbitrary base change. No inference from fibrewise isomorphisms is needed. An affine Grassmannian chart around the special pencil supplies a fixed complement to nearby R_C , identifying S_{R_C} , B , and E . There is no relation below degree d , so all lower neighbourhoods become constant. For a strict dominance transition, $\sum_{i,a} (\lambda'_{i,a})^2$ strictly increases; this sum is invariant under permutations of the decorated roots. The generic and special tuples therefore cannot be related by such a permutation. Part (1) separates their unmarked degree- d neighbourhoods. Flatness is asserted only for these finite neighbourhoods, not unrestricted failure schemes. $\square$

Corollary 8.2 (Exactly when the coarse data lose a labelled class). *The number of root-labelled congruence classes with the prescribed data is $\prod_i \# \mathcal{P}(m_i, c_i)$. It is one precisely when, for every i , either $c_i = 1$ or $m_i - c_i \leq 1$. The first possible ambiguity with fixed full determinant and all reduced rank loci occurs in dimension four, at $(m, c) = (4, 2)$, with partitions $(3, 1)$ and $(2, 2)$.*

Proof. Subtracting one from every part identifies $\mathcal{P}(m, c)$ with partitions of $m - c$ into at most c parts. There is one if $c = 1$ or $m - c \leq 1$. Otherwise $(m - c)$ and $(m - c - 1, 1)$ are distinct admissible partitions. The theorem gives the count and the minimal example. $\square$

8.1. The first relation as a standard graded Betti space. The finite invariant has a usual Hilbert-scheme interpretation. The first Betti space means the generator space in the first free module of a minimal graded resolution of the quotient, not the space of relations among generators of its defining ideal. The following statement retains the marked quotient presentation.

Proposition 8.3 (A graded Hilbert locus and its first Betti space). *In the fixed coordinates of Theorem 5.3, let $S = \text{Sym } H^*$, $N = \binom{n+1}{2}$, $M = \binom{N}{2}$, and $Q_R = S/((K_R) + S_{\geq d+1})$. Then*

$$\text{Tor}_1^S(Q_R, \mathbb{C})_d = K_R. \quad (8.5)$$

This identity commutes with arbitrary complex base change. The functor of graded quotients with Hilbert function

$$h_j = \begin{cases} \dim S_j, & 0 \leq j < d, \\ \dim S_d - M, & j = d, \\ 0, & j > d \end{cases}$$

is represented by $\text{Gr}(M, S_d)$. The pencil locus is a closed copy of $\text{Gr}(2, \text{Sym}^2 V)$. Its degree- d first Betti space recovers every scheme-theoretic pencil parameter, including parameters over nonreduced bases.

Proof. Put $J = (K_R) + S_{\geq d+1}$ and $S_+ = \bigoplus_{j>0} S_j$. The augmented sequence $0 \rightarrow J \rightarrow S \rightarrow Q_R \rightarrow 0$ gives $\text{Tor}_1^S(Q_R, \mathbb{C}) = J/S_+J$, since its map to S/S_+ is zero. Its degree- d part is K_R . The same argument over a base algebra applies to the degree- d subbundle with locally free quotient and commutes with pullback. A graded quotient with the indicated locally free components has zero ideal below degree d , a rank- M subbundle in degree d , and the full ideal in higher degrees. Conversely, every such subbundle defines an ideal, as its positive-degree products lie above degree d . This is the Grassmannian functor. The closed immersion and its inverse on the image are Theorem 5.3. This is not an inference from closed-point Torelli to unmarked deformation-groupoid equivalence. $\square$

9. A TOTALLY REAL RECIPROCAL LIKELIHOOD FIBRE

The spectral data also lead to a question independent of reconstruction: can every critical point of reciprocal likelihood on a definite real pencil be made real by a choice of real data? The assertion was formulated as Conjecture 4.5 in [40]. We give an explicit construction valid for every number of distinct eigenvalues and every multiplicity pattern. Its proof counts the critical points directly; it does not assume the complex likelihood-degree formula.

A real symmetric pencil containing a positive definite matrix can be written, after a real congruence and a choice of generators, as $\langle I, \text{diag}(a_1, \dots, a_n) \rangle$. Indeed, make the definite matrix the identity and orthogonally diagonalize the other symmetric matrix. Group its distinct eigenvalues as $\alpha_1 < \dots < \alpha_r$, with multiplicities $m_1, \dots, m_r > 0$ and $n = \sum_i m_i$. A two-dimensional pencil has $r \geq 2$. If σ_i is the sum of the data entries in the i -th eigenspace, the reciprocal score equations are the critical equations of

$$\tilde{\ell}(x, y) = -\sum_{i=1}^r m_i \log(\alpha_i x + y) - \sum_{i=1}^r \frac{\sigma_i}{\alpha_i x + y} \quad (9.1)$$

on the complement of $\prod_i (\alpha_i x + y) = 0$. The derivatives are rational and independent of the local branches of the logarithms. On the real complement one can equivalently use $\log |\alpha_i x + y|$.

Theorem 9.1 (Explicit totally real likelihood data). *Choose any real numbers*

$$\alpha_r < \beta_1 < \dots < \beta_{r-1}, \quad D(t) = \prod_{i=1}^r (t - \alpha_i), \quad Q(t) = \prod_{j=1}^{r-1} (t - \beta_j),$$

and prescribe

$$\sigma_i = \frac{Q(\alpha_i)}{D'(\alpha_i)}. \quad (9.2)$$

Then (9.1) has exactly $2r - 3$ critical points on its complex domain. Every one is real and nondegenerate. There is one critical point with $t = -y/x$ in each interval (α_i, α_{i+1}) , and one in each interval (β_j, β_{j+1}) ; there are no others. The same conclusion holds on a nonempty Euclidean open subset of the real data space. Thus the assertion of Conjecture 4.5 in [40] holds for all definite real pencils.

The data here are unrestricted real data, exactly as in that assertion. Neither positivity of the data matrix nor the assertion that all the critical points lie in the positive definite cone is part of the theorem. For example, the residues in (9.2) alternate in sign.

Lemma 9.2 (Two separated strings of roots). *With the notation of the theorem, put*

$$H(t) = \sum_{i=1}^r (n - m_i) \frac{D(t)}{t - \alpha_i}, \quad F(t) = nD(t)Q'(t) - H(t)Q(t). \quad (9.3)$$

The polynomial F has degree exactly $2r - 3$. It has a simple root in each of the intervals in the theorem, has no other root over $\mathbb{C}$, and is relatively prime to DQ .

Proof. Since $\sum_i (n - m_i) = n(r - 1)$, the terms of degree $2r - 2$ in F cancel. At a zero of D ,

$$F(\alpha_i) = -(n - m_i)D'(\alpha_i)Q(\alpha_i).$$

The factors $n - m_i$ are positive, all the values $Q(\alpha_i)$ have the same sign, and consecutive values $D'(\alpha_i)$ have opposite signs. Thus F changes sign in each of the $r - 1$ intervals between the α_i . At a zero of Q ,

$$F(\beta_j) = nD(\beta_j)Q'(\beta_j).$$

Here $D(\beta_j) > 0$, and consecutive values of Q' have opposite signs. There are therefore a further $r - 2$ roots in the intervals between the β_j . These intervals are pairwise disjoint and are disjoint

from all the former intervals. The upper bound $2r - 3$ on the degree forces exactly one root in each interval, all simple, and excludes any remaining real or complex root. The displayed endpoint values are nonzero, proving coprimality. This argument includes $r = 2$, when the second list of intervals is empty. $\square$

Proof of Theorem 9.1. Partial fractions give

$$\frac{Q(t)}{D(t)} = \sum_i \frac{\sigma_i}{t - \alpha_i}, \quad S(t) := \sum_i \frac{\sigma_i}{\alpha_i - t} = -\frac{Q(t)}{D(t)}. \quad (9.4)$$

On the chart $x \neq 0$, put $y = -xt$. Locally, up to a constant, the function becomes

$$-n \log x - \sum_i m_i \log(\alpha_i - t) - \frac{S(t)}{x}.$$

The first critical equation gives $x = S(t)/n$. If $S(t) = 0$ it cannot be satisfied on this chart. When $S(t) \neq 0$, the remaining critical equation is

$$\left. \frac{\partial \tilde{\ell}}{\partial t} \right|_{x=S(t)/n} = -\sum_i \frac{m_i}{t - \alpha_i} - n \frac{S'(t)}{S(t)} = -\frac{F(t)}{D(t)Q(t)} = 0. \quad (9.5)$$

The lemma now gives exactly $2r - 3$ solutions in this chart, with the real coordinates

$$x = -\frac{Q(t)}{nD(t)}, \quad y = -xt. \quad (9.6)$$

At each of these points $x \neq 0$ and $\alpha_i x + y \neq 0$, so none is an extraneous solution introduced by clearing denominators.

We must also exclude the omitted chart $x = 0$. Comparing the coefficients of t^{-1} and t^{-2} in (9.4) gives

$$\sum_i \sigma_i = 1, \quad \sum_i \alpha_i \sigma_i = \sum_i \alpha_i - \sum_j \beta_j.$$

At $x = 0$, the y -equation would imply $y = 1/n$. The x -equation would then require

$$\sum_i (n - m_i) \alpha_i = n \sum_j \beta_j. \quad (9.7)$$

But the left side is strictly smaller than $n(r - 1)\alpha_r$, whereas the right side is strictly larger. Thus there is no critical point on $x = 0$. This proves the assertion about the entire complex domain, not just the chosen coordinate chart.

For nondegeneracy, at a critical point in (x, t) -coordinates one has $\tilde{\ell}_{xx} = -n/x^2 \neq 0$. The derivative of the eliminated equation (9.5) is $-F'(t)/(D(t)Q(t)) \neq 0$. It is the Schur complement of $\tilde{\ell}_{xx}$ in the Hessian. Hence the Hessian is nonsingular. The coordinate change $(x, t) \mapsto (x, -xt)$ has nonzero Jacobian at these points, so nondegeneracy also holds in the original coordinates.

Finally allow Q to have any positive leading coefficient and $r - 1$ distinct real roots to the right of α_r . These polynomials form a nonempty Euclidean open set in their r -dimensional coefficient space. The linear map $Q \mapsto (Q(\alpha_i)/D'(\alpha_i))_{i=1}^r$ is an isomorphism, by polynomial interpolation. Scaling Q does not change its critical t -values and scales (x, y) by the same nonzero factor. Thus its image is the required open set of grouped data. The map from the original n data entries to their r group sums is surjective, so its inverse image is open and nonempty as well. $\square$

Example 9.3. For $(\alpha_1, \alpha_2, \alpha_3) = (1, 2, 3)$, $m_1 = m_2 = m_3 = 1$, and $(\beta_1, \beta_2) = (4, 5)$, the data are $(6, -6, 1)$ and

$$F(t) = 15t^3 - 130t^2 + 345t - 278.$$

Its three roots lie respectively in $(1, 2)$, $(2, 3)$, and $(4, 5)$. The theorem proves the same assertion for arbitrary r , not just for examples reached by root finding.

Remark 9.4 (Relation to the intrinsic invariant). The finite failure neighbourhood recovers the complex projective pencil and its spectral sheaf. To apply the real theorem one specifies a real structure and a definite member; these are not asserted to be recoverable from an unmarked complex scheme. After that specification, the construction uses only the ordered spectral points, their multiplicities, and the freely chosen β_j . The result is an independently stated consequence in reciprocal likelihood geometry, rather than a new name for Weierstrass–Segre classification. It proves the cited conjectural assertion without claiming that every real critical point is a statistically admissible maximum.

10. THE RECONSTRUCTED CRITICAL CORRESPONDENCE

This section identifies the algebraic object joining the finite inverse theorem to the real likelihood result. The reconstruction step is a functorial consequence of recovering the pencil. The additional content is a coordinate-free data construction and a relative critical algebra whose real points split into globally labelled sections.

10.1. The universal score scheme. For matrix formulas we regard quadrics as symmetric bilinear forms on a vector space; this amounts to dualizing the vector-space convention used earlier. Write $\mathcal{W} = \text{Sym}^2 V^*$. For a regular pencil $R \subset \mathcal{W}$, let $R^\times = \{K \in R : \det K \neq 0\}$. A data matrix S belongs to the same space $\mathcal{W}$. Define a section of the trivial bundle with fibre R^* on $R^\times \times \mathcal{W}$ by

$$\eta_{K,S}(H) = -\text{tr}(K^{-1}H) + \text{tr}(SK^{-1}HK^{-1}), \quad H \in R. \quad (10.1)$$

Its zero scheme $\mathfrak{C}_R$ is the reciprocal critical correspondence, with its projection to the data space $\mathcal{W}$. Over the reals these are the critical equations of $-\log |\det K| - \text{tr}(SK^{-1})$; the scheme itself uses no choice of logarithm and is defined over the complex field.

Theorem 10.1 (Reconstruction and base change of the critical correspondence). *Let $n \geq 3$ and $d = n^2 + 2n - 4$. For every regular pencil R , the abstract scheme $Z_R^{[d]}$ determines the isomorphism class of the correspondence*

$$\mathfrak{C}_R \longrightarrow \mathcal{W}$$

up to simultaneous congruence on the pencil and data space. On the regular pencil parameter scheme the score sections define a universal zero scheme. Its pullback along any complex base change is the corresponding relative critical scheme. The classifying map of a family in the framed first-relation locus therefore recovers this relative correspondence, including over nonreduced bases. No chosen data matrix or real form is part of the unmarked conclusion.

Proof. If g is invertible, put $K' = g^\dagger K g$, $H' = g^\dagger H g$, and $S' = g^\dagger S g$. Cyclic invariance of trace gives

$$\eta_{K',S'}(H') = \eta_{K,S}(H).$$

Thus simultaneous congruence identifies the score sections and their zero ideals, not just their closed points. By Theorem 5.1, an isomorphism of the finite neighbourhoods recovers the pencil up to projective congruence. A linear lift of that projective transformation supplies the displayed isomorphism of correspondences. Different lifts differ by a common scalar change of coordinates on the pencil and data space, so they leave its isomorphism class unchanged.

On $G^{\text{reg}} \subset \text{Gr}(2, \mathcal{W})$, let $\mathcal{R}$ be the tautological pencil bundle. On the determinant open in $\text{Tot } \mathcal{R}$, inversion of the universal form is a regular operation. Formula (10.1) is a section of the pullback of $\mathcal{R}^*$ after taking the product with $\mathcal{W}$. In a frame it has two entries. Their generated ideal is independent of the frame and its quotient commutes with arbitrary base change, as does

localization at the determinant. This proves the relative assertion directly, without an assertion that these score schemes are flat everywhere. Finally the closed immersion of Theorem 5.3 identifies a framed first-relation family with its classifying morphism to G . Pulling back the universal score construction proves the last assertion. It does not identify unmarked moduli stacks. $\square$

10.2. Spectral projectors and a congruence-covariant data section. Suppose now that A is real symmetric positive definite, B is real symmetric, and $R = \langle A, B \rangle$ is a pencil. Set $C = A^{-1}B$. Its distinct eigenvalues and multiplicities are

$$\alpha_1 < \cdots < \alpha_r, \quad m_1, \dots, m_r, \quad \sum_i m_i = n,$$

where $r \geq 2$. Its spectral idempotents are

$$P_i = \prod_{j \neq i} \frac{C - \alpha_j I}{\alpha_i - \alpha_j}, \quad P_i P_j = \delta_{ij} P_i, \quad \sum_i P_i = I. \quad (10.2)$$

They are selfadjoint for A . Define the grouped data map

$$\tau_C : \mathcal{W}_{\mathbb{R}} \longrightarrow \mathbb{R}^r, \quad \tau_C(S)_i = \text{tr}(P_i A^{-1} S). \quad (10.3)$$

For $K = xB + yA$, matrix functional calculus and cyclic trace give

$$-\log |\det K| - \text{tr}(SK^{-1}) = -\log \det A - \sum_i m_i \log |\alpha_i x + y| - \sum_i \frac{\tau_C(S)_i}{\alpha_i x + y}. \quad (10.4)$$

Thus the entire score scheme, not only its critical count, depends on S through the surjective linear map τ_C .

Choose $c > 0$ and $\alpha_r < \beta_1 < \cdots < \beta_{r-1}$, and set

$$\begin{aligned} D(t) &= \prod_i (t - \alpha_i), & Q(t) &= c \prod_j (t - \beta_j), \\ \sigma_i &= \frac{Q(\alpha_i)}{D'(\alpha_i)}, & S_0 &= A \sum_i \frac{\sigma_i}{m_i} P_i. \end{aligned} \quad (10.5)$$

The form S_0 is symmetric and $\tau_C(S_0) = \sigma$, since $\text{tr } P_i = m_i$. More generally the assignment $\sigma \mapsto A \sum_i (\sigma_i / m_i) P_i$ is a right inverse of τ_C ; all data with these grouped values are $S_0 + \ker \tau_C$. Under simultaneous congruence, $C \mapsto g^{-1} C g$, $P_i \mapsto g^{-1} P_i g$, and $S_0 \mapsto g^t S_0 g$. No eigenbasis has been chosen.

Theorem 10.2 (A totally real finite etale critical cover). *Fix the multiplicities $m_1, \dots, m_r$. On the locus of supplied real definite pairs (A, B) with these ordered multiplicities, there is a nonempty fibrewise open semialgebraic set of data defined by the following condition: the polynomial*

$$Q_S(t) = \sum_i \tau_C(S)_i \frac{D(t)}{t - \alpha_i} \quad (10.6)$$

has positive leading coefficient and $r - 1$ distinct real roots, all greater than α_r . On this set the reciprocal critical scheme has exactly $2r - 3$ points, all real and nondegenerate. They form globally labelled real analytic semialgebraic sections: $r - 1$ have their spectral coordinate in (α_i, α_{i+1}) , and $r - 2$ in the gaps between successive roots of Q_S .

In algebraic spectral coordinates the same family is the restriction of a finite etale algebra of rank $2r - 3$ over a specified Zariski open base. Its construction commutes with arbitrary base change on that base. The formula in (10.5) provides a congruence-covariant section of the data map, and its entire affine fibre has the same critical scheme.

Proof. Write $Q = Q_S$, and put

$$H(t) = \sum_i (n - m_i) \frac{D(t)}{t - \alpha_i}, \quad F(t) = nD(t)Q'(t) - H(t)Q(t). \quad (10.7)$$

Both apparent leading terms have degree $2r - 2$ and cancel. Writing $Q = c \prod_j (t - \beta_j)$, the coefficient of t^{2r-3} is

$$a = c \left(n \sum_j \beta_j - \sum_i (n - m_i) \alpha_i \right) > 0. \quad (10.8)$$

The strict inequality follows from $\beta_j > \alpha_r$, $n - m_i > 0$, and $\sum_i (n - m_i) = n(r - 1)$. Hence F has degree exactly $2r - 3$.

At α_i and β_j one has

$$F(\alpha_i) = -(n - m_i)D'(\alpha_i)Q(\alpha_i), \quad F(\beta_j) = nD(\beta_j)Q'(\beta_j).$$

The signs alternate along each string. The first string gives a root in every (α_i, α_{i+1}) , and the second gives one in every (β_j, β_{j+1}) . These $2r - 3$ disjoint intervals exhaust the degree. All roots are simple and F is coprime to DQ . This also treats $r = 2$, when the second string of intervals is empty.

We next prove the scheme statement and exclude the missing coordinate chart uniformly. Take the polynomial ring over $\mathbb{Q}$ in $\alpha_1, \dots, \alpha_r, \beta_1, \dots, \beta_{r-1}, c$, and localize at c , the products of distinct spectral and β -differences, a , $\text{disc}_t F$, and $\text{Res}_t(F, DQ)$. Call this ring $\mathcal{B}$. Every point of the separated real chamber lies in $\text{Spec } \mathcal{B}(\mathbb{R})$, by the preceding sign argument. For a linear F , its discriminant is understood as 1.

The partial fraction identity $Q/D = \sum_i \sigma_i/(t - \alpha_i)$ shows that

$$\sum_i \sigma_i = c, \quad \sum_i \alpha_i \sigma_i = c \left(\sum_i \alpha_i - \sum_j \beta_j \right).$$

On the chart $x = 0$ of the invertible-pencil locus the score in y forces $y = c/n$. Substitution in the score in x then forces $a = 0$. Since a is inverted, the quotient of the critical coordinate ring by (x) is zero. Thus x is a unit on the whole critical scheme, including nilpotent test rings; excluding this chart is not merely a pointwise argument.

Set $t = -y/x$ and $U(t) = \sum_i \sigma_i/(\alpha_i - t) = -Q(t)/D(t)$. In these coordinates the grouped function is $-n \log x - \sum_i m_i \log(\alpha_i - t) - U(t)/x$, with the logarithms used only to write its rational differential. The score in x gives $x = U/n$; the remaining score becomes $-F/(DQ)$. Consequently the critical coordinate algebra is

$$\mathcal{B}[t]/(F/a), \quad x = -\frac{Q(t)}{nD(t)}, \quad y = \frac{tQ(t)}{nD(t)}. \quad (10.9)$$

The resultant makes D and Q units in this algebra, so these formulas and the coordinate change are genuine inverse homomorphisms. The polynomial F/a is monic of degree $2r - 3$; its discriminant is a unit. The algebra is therefore finite free of that rank, and its derivative is a unit modulo the polynomial, which proves that it is étale. These algebraic identities persist under every base change of $\mathcal{B}$. The statement is local in spectral coordinates; no global flatness of the unrestricted critical correspondence is inferred.

At a critical point the second derivative in the eliminated x direction is $-n/x^2$. The complementary Schur factor is $-F'(t)/(D(t)Q(t))$. Both are nonzero, so the Hessian is nondegenerate. The change from (x, t) to (x, y) has Jacobian determinant $-x \neq 0$, preserving this conclusion. Each specified real interval contains a unique root. The implicit function theorem therefore supplies a real analytic root function locally, and uniqueness glues these functions globally with the stated

labels. Their graphs are semialgebraic, since they are specified by a polynomial equation and the interval inequalities. The ordered roots β_j of Q_S have the same local analytic and global semialgebraic property.

For fixed α , the map from the r coefficients of Q to σ is a linear isomorphism, with inverse (10.6). Polynomials with the specified simple separated roots and positive leading coefficient form a nonempty open semialgebraic coefficient set. Surjectivity and the explicit right inverse of τ_C make its preimage in the full data space nonempty and open. Spectral projectors give the same formulas along any real analytic family in the fixed multiplicity locus. Their congruence covariance was verified above; the kernel directions of τ_C do not occur in the score by (10.4). This completes all the assertions. $\square$

Corollary 10.3 (The finite-failure-to-critical construction). *An order- d finite failure invariant of a regular complex pencil, together with a supplied real definite realization and the spectral choices in (10.5), determines the above finite etale critical family up to simultaneous congruence. The all-real critical count and labelled real sections are independent of an eigenbasis. The real choices are additional inputs, not outputs of complex unmarked reconstruction.*

Proof. Apply Theorem 10.1, use the spectral sheaf to read the eigenvalues and their multiplicities in the supplied definite realization, and apply Theorem 10.2. Equations (10.2)–(10.5) verify invariance under all changes of eigenbasis and simultaneous congruence. The proof of total reality is an additional spectral argument, not a premise of the inverse theorem. $\square$

11. THE FINITE FLAT CRITICAL DIVISOR AND ITS RAMIFICATION

The critical cover of Section 10 was written on an affine spectral chart and away from its discriminant. We now retain the full projective critical divisor. It remains finite and flat when critical directions cross the omitted chart or collide. Its nonreduced fibres and its ramification are recovered from the same finite failure invariant. Thus the spectral construction carries more than the generic critical count or its real realization.

11.1. The projective score differential. Fix positive integers $m_1, \dots, m_r$, with $r \geq 2$ and $n = \sum_i m_i$. On the split semisimple locus write $K = xB + yA$, where A is nonsingular and $A^{-1}B$ has distinct spectral values α_i with these multiplicities. Over a nonreduced base, this means that mutually orthogonal idempotent projectors P_i of ranks m_i are supplied, with $\sum_i P_i = 1$ and $A^{-1}B = \sum_i \alpha_i P_i$. It is stronger than semisimplicity of the geometric fibres alone. The base field is $\mathbb{C}$; all the universal formulas below are already defined over $\mathbb{Q}$. Put

$$\begin{aligned} L_i &= \alpha_i x + y, & D_h &= \prod_i L_i, & D_i &= \prod_{j \neq i} L_j, \\ Q_h &= \sum_i \sigma_i D_i, & U(K) &= \text{tr}(SK^{-1}) = Q_h/D_h. \end{aligned} \tag{11.1}$$

Here $\sigma_i = \text{tr}(P_i A^{-1} S)$, with the projectors from (10.2). The polynomial Q_h is a binary form of degree $r - 1$; its affine leading coefficient is allowed to vanish.

Let $\mathcal{U}$ be the open subscheme of $\text{Spec } \mathbb{Q}[\alpha_1, \dots, \alpha_r, \sigma_1, \dots, \sigma_r]$ on which the α_i are pairwise distinct, Q_h has $r - 1$ distinct projective zeros, and these zeros are disjoint from those of D_h . Equivalently, invert the spectral differences, the binary discriminant of Q_h , and $\text{Res}(D_h, Q_h)$. The discriminant of a binary linear form is understood as 1; the resultant condition then also excludes the zero form. No discriminant of a score polynomial is inverted. In particular, this base allows critical-point collisions. All subsequent base changes are from $\mathcal{U}$.

Consider the following rational one-form on the projective line:

$$\begin{aligned}\omega &= \sum_i (n - m_i) d \log L_i - n d \log Q_h = \frac{\eta}{D_h Q_h}, \\ \eta &= Q_h \sum_i (n - m_i) D_i dL_i - n D_h dQ_h.\end{aligned}\tag{11.2}$$

Differentials are relative to the base. The equality $\sum_i (n - m_i) = n(r - 1)$ makes this expression independent of rescaling a representative (x, y) . If $\eta = \eta_x dx + \eta_y dy$, Euler's identity gives $x\eta_x + y\eta_y = 0$. The syzygy of (x, y) therefore gives a unique binary form G of degree $2r - 3$ such that

$$\eta = G(x, y)(x dy - y dx).\tag{11.3}$$

This argument is valid over the universal coefficient ring: reducing $x\eta_x + y\eta_y = 0$ modulo x first shows that x divides η_y , and then determines both coefficients. It remains valid after every base change.

On the chart $(x, y) = (1, -t)$, let $D(t) = \prod_i (t - \alpha_i)$ and $Q(t) = \sum_i \sigma_i D(t)/(t - \alpha_i)$. Direct substitution in (11.2) gives

$$G(1, -t) = -F(t), \quad F = nDQ' - Q \sum_i (n - m_i) \frac{D}{t - \alpha_i}.\tag{11.4}$$

Thus the form G , rather than the degree of a particular affine polynomial, records the whole critical divisor.

Theorem 11.1 (Finite flat reconstruction of the critical divisor). *Over $\mathcal{U}$, the zero scheme $\mathcal{Z} = V(G) \subset \mathbb{P}_{\mathcal{U}}^1$ is finite locally free of degree $2r - 3$, and is disjoint from $V(D_h Q_h)$. It is isomorphic, as a scheme over the base, to the entire reciprocal critical scheme on invertible pencil members. These assertions include nonreduced fibres and arbitrary base changes.*

Under a change of pencil frame the divisor transforms on the projective pencil line; under simultaneous congruence it is unchanged up to the induced isomorphism. For a family specified by its framed first relation spaces, with split semisimple spectral data and data matrices as above, the divisor is consequently reconstructed from its order- d failure neighbourhoods. Here $d = n^2 + 2n - 4$, and the supplied data and semisimple condition are additional to, not restrictions on, the all-pencil inverse theorem.

Proof. At a zero of L_i the residue of ω is $n - m_i$; at a zero of Q_h it is $-n$. These are nonzero units in characteristic zero. All the poles are simple and mutually disjoint on $\mathcal{U}$. Hence in every geometric fibre the numerator G is nonzero at every pole. This proves disjointness as schemes: a nonempty intersection would have a geometric point in some fibre. It also proves that G is not identically zero in any fibre.

Here is an elementary finite-flat argument which also deals with arbitrary, possibly nonreduced, bases. At each point of $\mathcal{U}$, choose a constant rational point of the projective line outside the finite zero set of its fibre of G . Such a point exists since the residue field has characteristic zero. After shrinking the base, the value of G there is a unit. A constant projective coordinate change takes this point to infinity. The scheme $\mathcal{Z}$ is then contained in the complementary affine chart, where its equation has unit leading coefficient and can be made monic of degree $2r - 3$. Its algebra is free on $1, t, \dots, t^{2r-4}$. These neighbourhoods cover the base, proving finite local freeness and the stated degree. The same presentations survive all base changes. They also identify $\mathcal{Z}$ as a relative effective Cartier divisor; compare the general criterion in [45, Tag 062Y].

We next prove the identification with the critical scheme, rather than only a bijection of closed points. Locally trivialize the projective direction by a nonsingular representative $K_0(t)$ and write

$K = \lambda K_0(t)$, where λ is invertible. Let $U_0(t) = \text{tr}(SK_0(t)^{-1})$. The rational differential of the reciprocal likelihood is obtained from

$$-n \log \lambda - \log \det K_0(t) - U_0(t)/\lambda.$$

Its radial score is $(-n\lambda + U_0)/\lambda^2$. Thus the critical coordinate ring imposes exactly $\lambda = U_0/n$, and makes U_0 a unit. Eliminating λ from the remaining score gives

$$-d \log \det K_0 - n d \log U_0 = \omega.$$

The determinant factorization and $U_0 = Q_h/D_h$ give the last equality. Since $\mathcal{Z}$ is disjoint from $D_h Q_h$, all divisions are by units on $\mathcal{Z}$. These are inverse coordinate-ring homomorphisms, including nilpotents. Globally the inverse sends a direction represented by K_0 to

$$K = \frac{U(K_0)}{n} K_0. \quad (11.5)$$

Replacing K_0 by aK_0 replaces $U(K_0)$ by $a^{-1}U(K_0)$, so this formula glues. It also proves that no projective chart has been omitted.

Both $-d \log \det K_0 - n d \log U(K_0)$ and (11.5) are invariant under pencil reparametrization. Simultaneous congruence preserves the trace and changes the determinant by a constant square; its relative logarithmic differential is therefore unchanged. Finally apply Theorem 5.3 and the universal construction in Theorem 10.1. This last reconstruction step uses a framed classifying family; it asserts neither an equivalence of unmarked moduli stacks nor recovery of a data matrix from an unmarked complex scheme alone. $\square$

11.2. Hessian degeneracy as scheme-theoretic ramification. Write $\pi : \mathcal{Z} \rightarrow \mathcal{U}$. The relative ramification scheme is defined by $\text{Fitt}_0 \Omega_{\mathcal{Z}/\mathcal{U}}$. The trace discriminant on the base is the determinant of $(a, b) \mapsto \text{Tr}_\pi(ab)$ on $\pi_* \mathcal{O}_{\mathcal{Z}}$; it is an ideal locally generated by a section of an invertible sheaf, not a choice of eigenbasis.

Theorem 11.2 (Ramification, Hessians, and the trace discriminant). *Under Theorem 11.1, the ramification ideal is exactly the ideal generated by the determinant of the reciprocal score Hessian on the critical scheme. The trace-discriminant ideal is locally generated, up to a unit, by the binary discriminant $\text{Disc}(G)$. Both ideals commute with arbitrary base change. The finite critical scheme is étale precisely off this trace discriminant, and the totally real cover of Theorem 10.2 is a restriction of this finite flat family.*

Proof. On a base neighbourhood with the monic presentation $\mathcal{A} = \mathcal{O}_{\mathcal{U}}[t]/(g(t))$ used above,

$$\Omega_{\mathcal{A}/\mathcal{O}_{\mathcal{U}}} = \mathcal{A} dt / (g'(t) dt).$$

Its zeroth Fitting ideal is (g') . In the local coordinates (λ, t) of the preceding proof the radial second derivative, after imposing the radial score, is $-n/\lambda^2$, a unit. The complementary Schur factor of the Hessian is the derivative of the eliminated score. That score is $v(t)g(t)$, with v a unit along the critical scheme, since its only denominators come from the disjoint poles. Its derivative modulo (g) is vg' . The Hessian determinant thus generates (g') . A change of smooth pencil coordinates multiplies this determinant on the critical scheme by the square of a Jacobian unit. This proves the claimed equality of ideal sheaves independently of the coordinates and without reducing the scheme.

In the basis $1, t, \dots, t^{m-1}$, $m = 2r - 3$, the determinant of the trace pairing for a monic polynomial algebra is $\text{disc}(g)$. One proof checks the Vandermonde identity for the polynomial with independent roots and then descends the resulting polynomial identity in its coefficients; consequently it remains valid for repeated roots and arbitrary coefficient rings. Making the leading coefficient a unit and changing projective frame multiply the binary discriminant by units. Hence

these local ideals are exactly $(\text{Disc}(G))$. This uses the classical trace-discriminant construction, not a new general discriminant theorem; see [45, Tags 0BVH, 0BJF].

The differential presentation, its Fitting ideal, and the matrix of the trace pairing all commute with base change. A monic algebra is étale exactly when g' is a unit in it, equivalently when its discriminant is a unit. On the separated real chamber the sign argument in Theorem 10.2 proves this condition and labels all m real roots. Its additional affine leading-coefficient localization merely selects one chart of the present family. No new sign argument is needed. $\square$

The trace discriminant on the base is not being identified with an arbitrarily scheme-theoretically imaged ramification locus: their multiplicities need not coincide. Likewise finite flatness is asserted on $\mathcal{U}$, not over data for which a zero of Q_h collides with a pole or becomes multiple. These exclusions concern the spectral data polynomial, not collisions of zeros of G .

11.3. A length-three critical fibre at a nonsingular pencil member.

Proposition 11.3 (A triple critical collision in a fixed pencil). *Let $A = I_3$, $B = \text{diag}(-1, 0, 1)$, and*

$$S_c = \text{diag}((1+c)/2, -c, (1+c)/2), \quad c \in \mathbb{C} \setminus \{0, -1\}.$$

The reciprocal critical schemes form a finite flat family of length three. At $c = 1$ all three critical points are real and reduced; one lies in the projective direction $x = 0$. At $c = -2/3$ the full critical scheme is supported at $K = I_3/3$ and is isomorphic to $\text{Spec } \mathbb{C}[u]/(u^3)$. The trace-discriminant ideal of this family is generated, up to a unit, by $(3c+2)^3$.

Proof. Here $D(t) = t(t^2 - 1)$, $Q(t) = t^2 + c$, and

$$F(t) = -(6c+4)t^2 + 2c. \quad (11.6)$$

The excluded parameters are exactly those where Q is repeated or meets a zero of D . Thus this entire family lies in $\mathcal{U}$. Homogenizing F to its projective degree three gives $\hat{F}(T, Z) = -(6c+4)T^2Z + 2cZ^3$. The associated projective direction is $[x : y] = [Z : -T]$. At $c = 1$ its zeros are $t = \pm 1/\sqrt{5}$ and the point $[T : Z] = [1 : 0]$, all simple. Formula (11.5) takes the latter to $I_3/3$.

At $c = -2/3$, $\hat{F} = -4Z^3/3$, so its whole zero scheme is in the chart $T = 1$, where $u = Z/T = -x/y$. The local equation is $-4u^3/3$. The radial reconstruction is an isomorphism on this nonreduced scheme and its reduced point is $I_3/3$, since $\text{tr } S_c = 1$. The binary cubic discriminant, or the trace calculation in a monic chart, is

$$\text{Disc}(\hat{F}) = 64c(3c+2)^3. \quad (11.7)$$

Since c is a unit on the stated base, this gives the claimed ideal. The special ramification ideal is (u^2) in $\mathbb{C}[u]/(u^3)$, in agreement with the Hessian theorem. $\square$

This example keeps the pencil and its spectral sheaf fixed. It shows why the critical correspondence must be retained with its scheme structure: the number of distinct critical points drops while the finite flat degree stays three. The order- d inverse theorem, followed by the supplied data family, reconstructs this collision and its trace-discriminant multiplicity. This is a structural consequence of the same reconstruction chain, not a replacement for the independent real-root argument or a claim that generic degree formulas are new.

ACKNOWLEDGMENT OF ASSISTANCE

AI assistance was used in developing and drafting the arguments and finite verification scripts. All structural proofs are supplied for independent mathematical scrutiny; computational receipts are not formal proof certificates. No AI system is listed as an author.

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